

|Python Inventory Analytics|

*[Inventory Analytics using Python & MySQL
Primary starting point: main.py]*

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Executive Summary

Python inventory analytics is an easy-to-use inventory visualisation software with a stylish dashboard for critical data insights. It is for small to medium-sized businesses or for personal inventory tracking.

The program is divided into 4 scripts; main.py, which calls all the other scripts, setup.py, which sets up the databases for the first time, called by database_handler.py, which handles all the queries and changes to the database, and finally visual_handler.py, which handles the GUI.

The program is first executed in the terminal, where it asks for login details of the mysql account it will be creating the database in, after which it loads the home page GUI.

In the home page, you can browse through the sample inventory and orders tables (shortened), or import your own after setting the inventory path and orders path, then importing and refreshing the screen.

On clicking modelling viewport, (as default it will show the options for the inventory table) you can construct a 2D or 3D (scatter/line) graph by repeatedly clicking the button to cycle through the options. Alternatively you can view the full table to browse through with all the table headings. For each item the application checks its basic inventory turnover value as well as if it has reached the preexisting inventory reorder point with a list of the same on the right hand side.

For orders, you can do much of the same except we can cycle through orders to see the amount spent and contact number of customers, with a list on the side to see the highest spenders in order.

main.py

[Source Code]

```
# main.py

import database_handler as database_handler

import visual_handler as visual_handler


database_handler.enter_mysql()

visual_handler.run_GUI()
```

[What the code does]

1. Imports a module named `database_handler` and aliases it to the same name.
2. Imports a module named `visual_handler` and aliases it to the same name.
3. Calls a function `enter_mysql()` from `database_handler` to establish a MySQL connection and prepare application data.
4. Calls a function `run_GUI()` from `visual_handler` to start a GUI loop.

[Runtime Architecture & Control Flow]

```
main.py
├── import database_handler
├── import visual_handler
├── database_handler.enter_mysql()    # DB setup (credentials &
connection)
└── visual_handler.run_GUI()         # GUI init + rendering & input
handling)
```

[Environment & Dependencies]

Not compatible with Python 3.13 (due to an error with X11).

[Third-party Libraries]

- MySQL: `mysql_connector_python` v8.4.0

- GUI: tkinter (preinstalled), matplotlib v3.10.3, ttkbootstrap v1.12.2
- File Extraction: numpy v2.2.6

database_handler.py

[Source Code]

```
# Requires MqSQL connector

import mysql.connector as conn
import setup as setup
import sys # to exit the program

conn_obj = "" # this is used to create the connection to mysql via
enter_mysql()

def enter_mysql():
    global conn_obj

    correct_credentials = False
    while correct_credentials != True:
        try:
            user_name = str(input("Please enter your MySQL username:
"))
            user_password = str(input("Please enter your MySQL
password: "))
            user_host = str(input("Please enter MySQL host:
('localhost' if using localhost): "))
            user_port = str(input("Please enter MySQL port: ('n' if
using localhost): "))
            print("Attempting connection...")

            if user_port == 'n':
                conn_obj = conn.connect(host = user_host, user =
user_name, passwd = user_password)
```

```

        else:
            conn_obj = conn.connect(host = user_host, port =
user_port, user = user_name, passwd = user_password)

            correct_credentials = True
    except Exception as e:
        exit_program = input("There was a mistake with the
username, password or port: " + str(e) + "\nPlease any key to try
again (or 'q' to quit) \n")
        if exit_program == "q":
            sys.exit(1) # exits program

    if conn_obj.is_connected() == True:
        print("Connected successfully")

    cursor = conn_obj.cursor()
    setup.initialise_tables(cursor)
    cursor.close()

def import_items(values):
    # values is a list of tuples; each tuple describes an item in the
format:
    (item_name,item_cost,item_gst,item_discount,item_final_cost,item_margi
n,item_stock,item_manufacturer_name,item_manufacturer_incharge,item_ma
nufacturer_contact_no)

    command = """INSERT INTO inventory
(item_name,item_cost,item_gst,item_discount,item_final_cost,item_margi

```

```
n,item_stock,item_restock_value,item_manufacturer_name,item_manufactur
er_incharge,item_manufacturer_contact_no) VALUES """
```

```
for i in values:
    command += str(i) + ","
```

```
command = command[:-1] + ";" # as last element ends with colon not
comma
```

```
cursor = conn_obj.cursor()
cursor.execute("USE main_database")
cursor.execute("DELETE FROM inventory")
cursor.execute("ALTER TABLE inventory AUTO_INCREMENT = 1")
cursor.execute(command)
cursor.execute("commit")
cursor.close()
```

```
def import_orders(values):
```

```
    # values is a list of tuples; each tuple describes an order in the
format:
```

```
(order_item_name,order_initial_cost,order_gst,order_discount,order_fin
al_cost,order_quantity,order_customer_name,order_customer_contact_no)
```

```
    command = """INSERT INTO orders
(order_item_name,order_date,order_initial_cost,order_gst,order_discoun
t,order_final_cost,order_quantity,order_customer_name,order_customer_c
ontact_no) VALUES """
```

```
for i in values:
    command += str(i) + ","
```



```
command = command[:-1] + ";" # as last element ends with colon not  
comma
```

```
cursor = conn_obj.cursor()  
cursor.execute("USE main_database")  
cursor.execute("DELETE FROM orders")  
cursor.execute("ALTER TABLE orders AUTO_INCREMENT = 1")  
cursor.execute(command)  
cursor.execute("commit")  
cursor.close()
```

```
def retrieve_via_sql_query(sql_select, sql_from, sql_where = "",  
sql_group_by = ""):
```

```
    command = "SELECT " + sql_select + "\n"
```

```
    if sql_where == "" and sql_group_by == "":
```

```
        # sql command with select and from only
```

```
        command += "FROM " + sql_from + ";" # if no where clause  
specified
```

```
    elif sql_where != "" and sql_group_by == "":
```

```
        #sql command with select, from and where
```

```
        command += "FROM " + sql_from + "\n"
```

```
        command += "WHERE " + sql_where + ";"
```

```
    else:
```

```
        #sql command with select, from and group by
```

```
        command += "FROM " + sql_from + "\n"
```

```
        command += "GROUP BY " + sql_group_by + ";"
```

```

        cursor = conn_obj.cursor()
        cursor.execute(command)

        data = cursor.fetchall()

        cursor.execute("commit")
        cursor.close()

    return(data)

def retrieve_headers(sql_from):
    command = "DESC " + sql_from

    cursor = conn_obj.cursor()
    cursor.execute(command)

    data = cursor.fetchall()

    cursor.execute("commit")
    cursor.close()

    return(data)

```

[Module Overview]

- Name: database_handler

- Purpose: Provides MySQL database connection establishment, initialization, data insertion into tables (inventory and orders), and retrieval operations (data rows and headers).
- Global Variables:
 - `conn_obj`: Initially an empty string, reassigned to hold the active MySQL connection object.

[Functions Documentation]

import_items(values):

- Input: `values` → list of tuples describing inventory items.
- Behavior:
 - Clears existing table contents (DELETE FROM inventory).
 - Resets auto-increment counter.
 - Inserts new values and commits.
- Side Effects: Executes SQL altering table contents.

import_orders(values)

- Input: `values` → list of tuples describing orders.
- Behavior:
 - Clears existing table contents and resets auto-increment.
 - Resets auto-increment counter.
 - Inserts new order rows and commits.
- Side Effects: Executes SQL altering table contents.

retrieve_via_sql_query(sql_select, sql_from, sql_where = "", sql_group_by = "")

- Input: Strings specifying SELECT, FROM, optional WHERE, and optional GROUP BY parts.
- Behavior: Dynamically constructs an SQL query and executes it.
- Output: Returns fetched rows (list of tuples).

- Side Effects: Executes SQL on the connected database.

retrieve_headers(sql_from)

- Input: Table name as a string.
- Behavior: Executes DESC <table> to fetch schema description.
- Output: Returns fetched description rows (list of tuples).
- Side Effects: Executes SQL on the connected database.

setup.py

[Source Code]

```
# Only run once on initialisation

def initialise_tables(cursor):
    # Working under databases called main_database and plot_database
    cursor.execute("CREATE DATABASE IF NOT EXISTS main_database")

    cursor.execute("USE main_database")

    # margin, discount is in percentages and number is in varchar to
handle leading zeros

    cursor.execute("DROP TABLE IF EXISTS inventory;") # Handles setup
being run more than once

    cursor.execute("""CREATE TABLE main_database.inventory
        (
            item_id int(10) not null AUTO_INCREMENT,
            item_name varchar(20) not null,
            item_cost float(8,2) not null,
            item_gst float(8,2) not null,
            item_discount float(5,2) not null,
            item_final_cost float(8,2) not null,
            item_margin float(5,2) not null,
            item_stock int(10) not null,
            item_restock_value int(10) not null,
            item_manufacturer_name varchar(40) not null,
            item_manufacturer_incharge varchar(20) not null,
```

```

        item_manufacturer_contact_no varchar(10) not
null,

        PRIMARY KEY (item_id)

    );""")

# Default data below

    cursor.execute("""INSERT INTO inventory
(item_name,item_cost,item_gst,item_discount,item_final_cost,item_margi
n,item_stock,item_restock_value,item_manufacturer_name,item_manufactur
er_incharge,item_manufacturer_contact_no)

VALUES

("Apples",139.00,10.00,5.00,122.55,50.00,500,50,"APPLE","Tim
Cook","0999666333"),

("Raspberry",139.00,10.00,5.00,122.55,50.00,500,50,"RASPBERRY
PI","Eben Upton","0111222333"),

("Orange",139.00,10.00,5.00,122.55,50.00,500,50,"ORANGE PI","Zhao
Yifan","0555666777"),

("Bannana",139.00,10.00,5.00,122.55,50.00,500,50,"BANNANA PI","U.N.
Owen","0333444555"),

        ("Potato",139.00,10.00,5.00,122.55,50.00,500,50,"LE
POTATO LIBRE","W.H. Onos","0777888999");

    """)

    cursor.execute('commit')

# discount is in percentages and number is in varchar to handle
leading zeros

```

```
cursor.execute("DROP TABLE IF EXISTS orders;") # Handles setup being  
run more than once
```

```
cursor.execute("""CREATE TABLE main_database.orders  
(  
    order_id int(10) not null AUTO_INCREMENT,  
    order_item_name varchar(20) not null,  
    order_date date not null,  
    order_initial_cost float(8,2) not null,  
    order_gst float(8,2) not null,  
    order_discount float(5,2) not null,  
    order_final_cost float(8,2) not null,  
    order_quantity int(10) not null,  
    order_customer_name varchar(20) not null,  
    order_customer_contact_no varchar(10),  
    PRIMARY KEY (order_id)  
);""")
```

Default data below

```
cursor.execute("""INSERT INTO orders  
(order_item_name,order_date,order_initial_cost,order_gst,order_discount,  
order_final_cost,order_quantity,order_customer_name,order_customer_c  
ontact_no)  
VALUES  
("Apples","2025-04-08",139.00,10.00,5.00,122.55,1,"Tom  
Bakes","1999666333"),  
  
("Raspberry","2025-04-08",139.00,10.00,5.00,122.55,1,"Nebe  
Downton","1111222333"),
```

```

("Orange", "2025-04-08", 139.00, 10.00, 5.00, 122.55, 1, "Zhao
Yistan", "1555666777"),

("Bannana", "2025-04-08", 139.00, 10.00, 5.00, 122.55, 1, "K.N.
Owen", "1333444555"),

("Potato", "2025-04-08", 139.00, 10.00, 5.00, 122.55, 1, "W.E.
Noe", "1777888999");

        """
    cursor.execute('commit')

```

[Module Overview]

- Name: setup
- Purpose: Defines and initialise_tables, which ensures the database main_database and its tables (inventory, orders) exist, and populates them with default rows.

[Function Documentation]

initialise_tables(cursor)

- Input:
 - cursor: Database cursor object.
- Behavior:
 1. Creates main_database if it does not exist.
 2. Switches to main_database.
 3. Drops and recreates inventory with specified columns and constraints.
 4. Inserts multiple default rows into inventory.
 5. Commits transaction.
 6. Drops and recreates orders with specified columns and constraints.
 7. Inserts multiple default rows into orders.

8. Commits transaction.

- Output: None (no explicit return).
- Side Effects: Alters database schema and inserts rows

visual_handler.py

[Source Code]

```
import tkinter as tk # for general GUI
from tkinter import ttk # for access to the scrollbar and treeview
widgets

from tkinter.filedialog import askopenfilename # for inputting the CSV
file

from numpy import genfromtxt # for handling the CSV

import database_handler as database_handler # for handling the
database

import matplotlib.pyplot as plt # for plotting

from ttkbootstrap.widgets import Button, Treeview, Label # for
creating come stylised widgets

from ttkbootstrap import Style # for overall colour pallate of the
widgets

import datetime # for retreiving date

import sys # for quitting the application


def run_GUI():
    # all window definitions
    root = tk.Tk()
    root.title("Python Inventory Analytics")
    root.geometry("660x365")
    root.protocol('WM_DELETE_WINDOW', sys.exit)


    home_view = tk.Frame(root)
    home_view.grid(row = 1, column = 1, sticky = 'news')
```

```

modelling_view = tk.Frame(root)
modelling_view.columnconfigure(0,weight = 1)
modelling_view.rowconfigure(1, weight = 1)
modelling_view.grid(row = 1, column = 1, sticky = 'news')

# setting style
style = Style("solar")

# functions for home view
def populate_data():
    # removing response message if any
    response_message.set("")

# inventory viewer
inventory_data =
database_handler.retrieve_via_sql_query("item_id,item_name,item_cost,i
tem_final_cost,item_stock","inventory")
inventory_viewer = Treeview(inventory_tab,
                            columns =
("item_id","item_name","item_cost","item_final_cost","item_stock"),
                            show = 'headings',
                            height = 13,
                            bootstyle = 'success'
                            )
inventory_viewer.grid(row = 1,
                    column = 0,
                    sticky = "e"
                    )

```

```

# creating the scrollbar

scrollbar = ttk.Scrollbar(inventory_tab, orient = "vertical",
command = inventory_viewer.yview)

scrollbar.grid(row = 1,
               column = 1,
               sticky = "nsew"
               )

inventory_viewer.configure(yscrollcommand = scrollbar.set)

# initialising columns

inventory_viewer.column("item_id", anchor = "center", width = 45)
inventory_viewer.heading('item_id', text = 'S.No')

inventory_viewer.column("item_name", anchor = "center", width = 75)
inventory_viewer.heading('item_name', text = 'Name')

inventory_viewer.column("item_cost", anchor = "center", width = 55)
inventory_viewer.heading('item_cost', text = 'Cost')

inventory_viewer.column("item_final_cost", anchor = "center", width
= 55)
inventory_viewer.heading('item_final_cost', text = 'Total')

inventory_viewer.column("item_stock", anchor = "center", width = 50)
inventory_viewer.heading('item_stock', text = 'Stock')

inventory_viewer.grid(row = 1, column = 0)

```

```

# insert values into inventory_viewer

for i in inventory_data:

    inventory_viewer.insert(parent = '', index = tk.END, values = i)


# orders viewer code

orders_data =
database_handler.retrieve_via_sql_query("order_id,order_item_name,order_customer_name,order_final_cost,order_quantity","orders")


orders_viewer = Treeview(orders_tab,

                        columns =
("order_id","order_item_name","order_customer_name","order_final_cost"
,"order_quantity"),

                        show = 'headings',
                        height = 13,
                        bootstyle = 'success'
                        )

orders_viewer.grid(row = 1,

                  column = 0,
                  sticky = "e"
                  )


# creating the scrollbar

scrollbar = ttk.Scrollbar(orders_tab, orient = "vertical", command =
orders_viewer.yview)

scrollbar.grid(row = 1,

              column = 1,
              sticky = "nsew"
              )

```

```

orders_viewer.configure(yscrollcommand = scrollbar.set)

# initialising columns
orders_viewer.column("order_id", anchor = "center", width = 45)
orders_viewer.heading('order_id', text = 'S.No')

orders_viewer.column("order_item_name", anchor = "center", width =
75)
orders_viewer.heading('order_item_name', text = 'Item')

orders_viewer.column("order_customer_name", anchor = "center", width
= 60)
orders_viewer.heading('order_customer_name', text = 'Name')

orders_viewer.column("order_final_cost", anchor = "center", width =
55)
orders_viewer.heading('order_final_cost', text = 'Total')

orders_viewer.column("order_quantity", anchor = "center", width =
45)
orders_viewer.heading('order_quantity', text = 'Amt')

orders_viewer.grid(row = 1, column = 0)

# insert values into orders_viewer
for i in orders_data:
    orders_viewer.insert(parent = '', index = tk.END, values = i)

def import_database():

```

```

    if full_inventory_path.get() != "" and full_orders_path.get() !=
"":
        try:
            items = genfromtxt(full_inventory_path.get(), delimiter = ",",
dtype = None, skip_header = 1, encoding = "utf8")
            database_handler.import_items(items)
        except Exception as e:
            # error with csv file
            print(e)
            response_message.set("Import Unsuccessful; Please check your
CSV")
            spacer.config(fg = "red")
            return None

        try:
            orders = genfromtxt(full_orders_path.get(), delimiter = ",",
dtype = None, skip_header = 1, encoding = "utf8")
            database_handler.import_orders(orders)
        except:
            # error with csv file
            response_message.set("Import Unsuccessful; Please check your
CSV")
            spacer.config(fg = "red")
            return None

        # if function reaches here, the code was sucessfull
        response_message.set("Import Successful; Please Refresh")
        spacer.config(fg = "green")
    else:

```

```

        response_message.set("Import Unsuccessful; Please try again")
        spacer.config(fg = "red")

def set_inventory_path():
    full_inventory_path.set(tk.filedialog.askopenfilename())
    temp = full_inventory_path.get()
    inventory_path.set("Inventory path: \n" + temp[:20] + "...")

def set_orders_path():
    full_orders_path.set(tk.filedialog.askopenfilename())
    temp = full_inventory_path.get()
    orders_path.set("Orders path: \n" + temp[:20] + "...")

def refresh():
    populate_data()
    populate_inventory_low_stocks_data()
    populate_orders_highest_spends_data()
    set_inventory_response_message("Database refreshed")

def switch_to_modelling_view():
    modelling_view.tkraise()

# home view GUI
database_label = tk.Label(home_view, text = "Database Viewer",
    relief = "ridge", font = "TkFixedFont")
database_label.grid(row = 0,
                    column = 0,
                    padx = 10,
                    pady = 10,

```



```

        sticky = "nesw"
    )

    other_functions_label = tk.Label(home_view, text = ' 🏠 📁 📁 📁 Other
Functions 📁 📁 📁 📁 ', relief = "ridge", font = "TkFixedFont")
    other_functions_label.grid(row = 0,
                              column = 1,
                              sticky = "ew"
                              )

    button_frame = tk.Frame(home_view)
    button_frame.grid(row = 1,
                      column = 1,
                      sticky = "nsew"
                      )

    first_row_frame = tk.Frame(button_frame)
    first_row_frame.grid(row = 0,
                         columnspan = 3,
                         pady = 5,
                         sticky = "nsew"
                         )

    modelling_view_button = Button(first_row_frame,
                                   text = "📁 📁 📁 📁 \n 📊 Modelling \n
Viewport \n 📁 📁 📁 📁",
                                   command = switch_to_modelling_view,
                                   bootstyle = "primary-outline"
                                   )

```

```

modelling_view_button.grid(row = 0,
                            column = 0,
                            pady = 5,
                            sticky = "e"
                            )

refresh_database_button = Button(first_row_frame,
                                text = "Database \n 🔄 Refresh \n Database \n ",
                                command = refresh,
                                bootstyle = "warning-outline"
                                )

refresh_database_button.grid(row = 0,
                            column = 1,
                            padx = 5,
                            pady = 5,
                            sticky = "nsew"
                            )

full_inventory_path = tk.StringVar()
full_orders_path = tk.StringVar()

import_database_button = Button(first_row_frame,
                                text = "Database \n 📄 Import \n Database \n ",
                                command = import_database,

```

```

        bootstyle = "success-outline"
    )

import_database_button.grid(row = 0,
                             column = 2
                             )

inventory_path = tk.StringVar()
inventory_path_button = tk.Button(button_frame,
                                   textvariable = inventory_path,
                                   command = set_inventory_path,
                                   font = "TkFixedFont",
                                   height = 3
                                   )

inventory_path.set("Set inventory path")

inventory_path_button.grid(row = 1,
                           colspan = 3,
                           sticky = "nsew"
                           )

orders_path = tk.StringVar()
orders_path_button = tk.Button(button_frame,
                                textvariable = orders_path,
                                command = set_orders_path,
                                font = "TkFixedFont",
                                height = 3
                                )

```

```

orders_path.set("Set orders path")

orders_path_button.grid(row = 2,
                        colspan = 3,
                        sticky = "nsew"
                        )

response_message = tk.StringVar()

spacer = tk.Label(button_frame, textvariable = response_message,
relief = "raised")

spacer.grid(row = 3,
            colspan = 3,
            pady = 5,
            sticky = "nsew"
            )

response_message.set("")

exit_button = tk.Button(button_frame,
                        text = "##### \nQuit
Python-Inventory-Analytics\n #####",
                        command = sys.exit,
                        font = "TkFixedFont",
                        height = 2
                        )

exit_button.grid(row = 4,
                colspan = 3,
                sticky = "ew"
                )

```

```

# Notebook to contain the two tables in our database

table_frame = tk.Frame(home_view)

table_frame.grid(row = 1,
                  column = 0,
                  sticky = "nsew"
                  )

# Each column must take equal amount of space

table_frame.rowconfigure(0, weight = 1)
table_frame.rowconfigure(1, weight = 1)
table_frame.rowconfigure(2, weight = 1)

tables_notebook = ttk.Notebook(table_frame)
tables_notebook.grid(row = 1,
                     padx = 10
                     )

inventory_tab = tk.Frame(tables_notebook)
tables_notebook.add(inventory_tab, text = "    Inventory    ")

orders_tab = tk.Frame(tables_notebook)
tables_notebook.add(orders_tab, text = "    Orders    ")

populate_data() # populates data in the above notebook

# functions for modelling view GUI
def switch_to_home_view():
    home_view.tkraise()

```

```

def switch_to_inventory_models_view():
    inventory_models_view.tkraise()

def switch_to_order_models_view():
    order_models_view.tkraise()

def set_inventory_response_message(new_message):
    previous_text = inventory_response_message.get()
    new_text = ""
    if previous_text == "          Action status will be displayed here:
\n\n":
        # ie. response msg board is empty
        new_text = previous_text[:-1] + "          " + new_message + "          "
\n"
    elif previous_text.count(" ") == 2:
        # if one msg already in msg board
        new_text = previous_text + "          " + new_message + "          "
    elif previous_text.count(" ") == 4:
        # message board full
        messages = previous_text.split(" ")
        messages[1] = messages[3]
        messages[3] = new_message
        new_text = " ".join(messages)

    inventory_response_message.set(new_text)

def set_orders_response_message(new_message):
    previous_text = orders_response_message.get()

```

```

new_text = ""

if previous_text == "           Action status will be displayed here:
\n\n":
    # ie. response msg board is empty
    new_text = previous_text[:-1] + "           ▪ " + new_message + " ▪
\n"

elif previous_text.count("▪") == 2:
    # if one msg already in msg board
    new_text = previous_text + "           ▪ " + new_message + " ▪ "

elif previous_text.count("▪") == 4:
    # message board full
    messages = previous_text.split(" ▪ ")
    messages[1] = messages[3]
    messages[3] = new_message
    new_text = " ▪ ".join(messages)

orders_response_message.set(new_text)

def close_plot():
    set_inventory_response_message("All open plots closed")
    set_orders_response_message("All open plots closed")
    plt.close()

def draw_plot(use_orders_table = False):
    #first, remove existing plot
    close_plot()

    if use_orders_table == False:
        x_axis = x_axis_column_name.get()

```

```

y_axis = y_axis_column_name.get()
z_axis = z_axis_column_name.get()
else:
    x_axis = orders_x_axis_column_name.get()
    y_axis = orders_y_axis_column_name.get()
    z_axis = orders_z_axis_column_name.get()

# without z-axis
if x_axis != "unspecified" and y_axis != "unspecified" and z_axis
== "unspecified":
    if use_orders_table == False:
        retrieved_data =
database_handler.retrieve_via_sql_query(str(x_axis + "," + y_axis),
"inventory")
    else:
        retrieved_data =
database_handler.retrieve_via_sql_query(str(x_axis + "," + y_axis),
"orders")
    x_axis_list = [retrieved_data[i][0] for i in
range(0,len(retrieved_data))]
    y_axis_list = [retrieved_data[i][1] for i in
range(0,len(retrieved_data))]

    ax = plt.axes()
    ax.plot(x_axis_list, y_axis_list, marker = 'x')
    ax.set_title(x_axis + " vs " + y_axis)
    ax.set_xlabel(x_axis)
    ax.set_ylabel(y_axis)

    set_inventory_response_message("x-y Graph drawn")

```



```

plt.show()

# with z-axis

elif x_axis != "unspecified" and y_axis != "unspecified" and z_axis
!= "unspecified":

    if use_orders_table == False:

        retrieved_data =
database_handler.retrieve_via_sql_query(str(x_axis + "," + y_axis +
"," + z_axis), "inventory")

    else:

        retrieved_data =
database_handler.retrieve_via_sql_query(str(x_axis + "," + y_axis +
"," + z_axis), "orders")

    x_axis_list = [retrieved_data[i][0] for i in
range(0,len(retrieved_data))]

    y_axis_list = [retrieved_data[i][1] for i in
range(0,len(retrieved_data))]

    z_axis_list = [retrieved_data[i][2] for i in
range(0,len(retrieved_data))]

    # checking if type is string as need to treat them diffenently in
plot

    x_axis_type_string = False
    y_axis_type_string = False
    z_axis_type_string = False

    if isinstance(x_axis_list[0], int) == False and
isinstance(x_axis_list[0], float) == False:

        # to check if string/date

        x_axis_type_string = True

```

```

        x_plot_data = range(len(x_axis_list))
    else:
        x_plot_data = x_axis_list

    if isinstance(y_axis_list[0], int) == False and
isinstance(y_axis_list[0], float) == False:
        #to check if string/date
        y_axis_type_string = True
        y_plot_data = range(len(y_axis_list))
    else:
        y_plot_data = y_axis_list

    if isinstance(z_axis_list[0], int) == False and
isinstance(z_axis_list[0], float) == False:
        #to check if string/date
        z_axis_type_string = True
        z_plot_data = range(len(z_axis_list))
    else:
        z_plot_data = z_axis_list

ax = plt.axes(projection='3d')

if plot_type.get()[17:] == "Scatter":
    set_orders_response_message("x-y-z Scatter Graph drawn")
    ax.scatter(x_plot_data, y_plot_data, z_plot_data, c=
range(len(z_axis_list)), cmap='plasma', marker='x') # colours require
numeric data always
if plot_type.get()[17:] == "Line":
    set_orders_response_message("x-y-z Line Graph drawn")

```

```

ax.plot3D(x_plot_data, y_plot_data, z_plot_data)

# replacing the pseudo-numbers (for string data) in the above
statement by actual data if present (required to avoid datatype
issues)

if x_axis_type_string == True:
    ax.set(xticks=range(len(x_axis_list)), xticklabels=x_axis_list)

if y_axis_type_string == True:
    ax.set(yticks=range(len(y_axis_list)), yticklabels=y_axis_list)

if z_axis_type_string == True:
    ax.set(zticks=range(len(z_axis_list)), zticklabels=z_axis_list)

ax.set_title(x_axis + " vs " + y_axis + " vs " + z_axis)
ax.set_xlabel(x_axis, labelpad=20)
ax.set_ylabel(y_axis, labelpad=20)
ax.set_zlabel(z_axis, labelpad=20)

plt.show()

def toggle_plot_type():
    plot_type_list = ["Scatter", "Line"]
    current_index = plot_type_list.index(plot_type.get()[17:])

    current_index += 1
    current_index = current_index % len(plot_type_list)

```

```

if current_index == 0:
    set_inventory_response_message("Graph type set to Scatter")
    set_orders_response_message("Graph type set to Scatter")
else:
    set_inventory_response_message("Graph type set to Line")
    set_orders_response_message("Graph type set to Line")

plot_type.set("📊 3D Graph type: " +
plot_type_list[current_index])

def set_x_axis(use_orders_table = False):
    # retriving the column list
    if use_orders_table == False:
        table_information =
database_handler.retrieve_headers("inventory")

        column_list = []
        for i in range(0,len(table_information)):
            column_list.append(table_information[i][0])

        # finding current index
        current_column = x_axis_column_name.get()
        try:
            current_column_index = column_list.index(current_column)
        except:
            #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)
            current_column_index = -1

```

```

#in case at the last column

if current_column_index + 1 == len(column_list):
    current_column_index = -1

x_axis_column_name.set(column_list[current_column_index + 1])

#formatting the text

if len(x_axis_column_name.get()) < 25:
    x_axis.set("Set Graph's X-Axis:\n" + x_axis_column_name.get())
else:
    x_axis.set("Set Graph's X-Axis:\n" +
x_axis_column_name.get()[0:22] + "...")
else:
    # use orders information

    table_information = database_handler.retrieve_headers("orders")

    column_list = []
    for i in range(0,len(table_information)):
        column_list.append(table_information[i][0])

    # finding current index

    current_column = orders_x_axis_column_name.get()
    try:
        current_column_index = column_list.index(current_column)
    except:
        #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)

        current_column_index = -1

```

```

#in case at the last column

if current_column_index + 1 == len(column_list):
    current_column_index = -1

orders_x_axis_column_name.set(column_list[current_column_index +
1])

#formatting the text

if len(orders_x_axis_column_name.get()) < 25:
    orders_x_axis.set("Set Graph's X-Axis:\n" +
orders_x_axis_column_name.get())
else:
    orders_x_axis.set("Set Graph's X-Axis:\n" +
orders_x_axis_column_name.get()[0:22] + "...")

def set_y_axis(use_orders_table = False):
    # retriving the column list

    if use_orders_table == False:
        table_information =
database_handler.retrieve_headers("inventory")
        column_list = []
        for i in range(0,len(table_information)):
            column_list.append(table_information[i][0])

    # finding current index
    current_column = y_axis_column_name.get()
    try:
        current_column_index = column_list.index(current_column)
    except:

```

```

        #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)

        current_column_index = -1

    #in case at the last column
    if current_column_index + 1 == len(column_list):
        current_column_index = -1

    y_axis_column_name.set(column_list[current_column_index + 1])

    #formatting the text
    if len(y_axis_column_name.get()) < 25:
        y_axis.set("Set Graph's Y-Axis:\n" + y_axis_column_name.get())
    else:
        y_axis.set("Set Graph's Y-Axis:\n" +
y_axis_column_name.get()[0:22] + "...")
    else:
        table_information = database_handler.retrieve_headers("orders")
        column_list = []
        for i in range(0,len(table_information)):
            column_list.append(table_information[i][0])

    # finding current index
    current_column = orders_y_axis_column_name.get()
    try:
        current_column_index = column_list.index(current_column)
    except:
        #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)

```

```

        current_column_index = -1

        #in case at the last column
        if current_column_index + 1 == len(column_list):
            current_column_index = -1

        orders_y_axis_column_name.set(column_list[current_column_index +
1])

        #formatting the text
        if len(orders_y_axis_column_name.get()) < 25:
            orders_y_axis.set("Set Graph's Y-Axis:\n" +
orders_y_axis_column_name.get())
        else:
            orders_y_axis.set("Set Graph's Y-Axis:\n" +
orders_y_axis_column_name.get()[0:22] + "...")

    def set_z_axis(use_orders_table = False):
        if use_orders_table == False:
            # retriving the column list
            table_information =
database_handler.retrieve_headers("inventory")
            column_list = []
            for i in range(0,len(table_information)):
                column_list.append(table_information[i][0])

            #as 3d plotting is optional,
            column_list.append("unspecified")

```



```

# finding current index
current_column = z_axis_column_name.get()
try:
    current_column_index = column_list.index(current_column)
except:
    #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)
    current_column_index = -1

#in case at the last column
if current_column_index + 1 == len(column_list):
    current_column_index = -1

z_axis_column_name.set(column_list[current_column_index + 1])

#formatting the text
if len(z_axis_column_name.get()) < 25:
    z_axis.set("Set Graph's Z-Axis:\n" + z_axis_column_name.get())
else:
    z_axis.set("Set Graph's Z-Axis:\n" +
z_axis_column_name.get()[0:22] + "...")
else:
    # retriving the column list
    table_information = database_handler.retrieve_headers("orders")
    column_list = []
    for i in range(0,len(table_information)):
        column_list.append(table_information[i][0])

#as 3d plotting is optional,

```

```

column_list.append("unspecified")

# finding current index
current_column = orders_z_axis_column_name.get()
try:
    current_column_index = column_list.index(current_column)
except:
    #if unspecified, just sets it to -1 then +1 while displaying so
sets to the zeroth position)
    current_column_index = -1

#in case at the last column
if current_column_index + 1 == len(column_list):
    current_column_index = -1

orders_z_axis_column_name.set(column_list[current_column_index +
1])

#formatting the text
if len(orders_z_axis_column_name.get()) < 25:
    orders_z_axis.set("Set Graph's Z-Axis:\n" +
orders_z_axis_column_name.get())
else:
    orders_z_axis.set("Set Graph's Z-Axis:\n" +
orders_z_axis_column_name.get()[0:22] + "...")

def tabularise_full_inventory_database(use_orders_table = False):
    if use_orders_table == False:
        # Create a new window

```

```

full_inventory_database_window =
tk.Toplevel(inventory_models_view)

full_inventory_database_window.title("Full inventory viewer")

full_inventory_database_window.geometry("1757x360")

full_inventory_database_window.resizable(False, False)

# Get header list

headers = database_handler.retrieve_headers("inventory")

column_list = []

for i in range(0, len(headers)):

    header_name = headers[i][0][5:]

    if "manufacturer_" in header_name:

        header_name = "Manf. " + header_name[13:]

    column_list.append(header_name)

full_data =
database_handler.retrieve_via_sql_query("*", "inventory")

full_inventory_database_table =
Treeview(full_inventory_database_window,

          columns = column_list,

          show = 'headings',

          height = 17,

          bootstyle = 'success'

          )

full_inventory_database_table.grid(row = 0,

                                   column = 0,

                                   sticky = "nsew"

                                   )

```

```

    # creating the scrollbar

    scrollbar = ttk.Scrollbar(full_inventory_database_window, orient
= "vertical", command = full_inventory_database_table.yview)

    scrollbar.grid(row = 0,
                    column = 1,
                    sticky = "nsew"
                    )

    full_inventory_database_table.configure(yscrollcommand =
scrollbar.set)

    # initialising columns
    for i in column_list:
        full_inventory_database_table.column(i, anchor = "center",
width = 145)
        full_inventory_database_table.heading(i, text = i)

    # insert values into full_inventory_database_window
    for i in full_data:
        full_inventory_database_table.insert(parent = '', index =
tk.END, values = i)

    set_inventory_response_message("Opened full inventory viewer")
else:
    # Create a new window
    full_orders_database_window = tk.Toplevel(order_models_view)
    full_orders_database_window.title("Full Orders viewer")
    full_orders_database_window.geometry("1465x360")
    full_orders_database_window.resizable(False, False)

```

```

# Get header List

headers = database_handler.retrieve_headers("orders")

column_list = []

for i in range(0, len(headers)):
    header_name = headers[i][0][6:]
    column_list.append(header_name)

full_data = database_handler.retrieve_via_sql_query("*", "orders")

full_orders_database_table =
Treeview(full_orders_database_window,
          columns = column_list,
          show = 'headings',
          height = 17,
          bootstyle = 'success'
          )

full_orders_database_table.grid(row = 0,
                                column = 0,
                                sticky = "nsew"
                                )

# creating the scrollbar

scrollbar = ttk.Scrollbar(full_orders_database_window, orient =
"vertical", command = full_orders_database_table.yview)

scrollbar.grid(row = 0,
               column = 1,
               sticky = "nsew"
               )

```

```

    full_orders_database_table.configure(yscrollcommand =
scrollbar.set)

    # initialising columns
    for i in column_list:
        full_orders_database_table.column(i, anchor = "center", width =
145)

        full_orders_database_table.heading(i, text = i)

    # insert values into full_inventory_database_window
    for i in full_data:
        full_orders_database_table.insert(parent = '', index = tk.END,
values = i)

        set_orders_response_message("Opened full orders viewer")

def set_selected_item_name():
    # retriving the name list
    table_information =
database_handler.retrieve_via_sql_query("item_id,item_name","inventory
")

    # finding current index
    current_column_index = int(selected_item_id.get()[8:])

    # setting to name at the id
    set_inventory_response_message("Item set to: " +
table_information[current_column_index][1] )

    selected_item_name.set("Item name: " +
table_information[current_column_index][1])

```

```

def select_previous_item():
    # retriving the name list
    table_information =
database_handler.retrieve_via_sql_query("item_id,item_name","inventory
")

    # finding current index
    current_column_index = int(selected_item_id.get()[8:])

    #in case at the first column
    if current_column_index == 0:
        current_column_index = len(table_information) - 1
    else:
        current_column_index = current_column_index - 1

    selected_item_id.set("Item id: " +
str(table_information[current_column_index][0]-1))
    set_selected_item_name()
    set_basic_inventory_turnover()
    set_reorder_warning()

def select_next_item():
    # retriving the name list
    table_information =
database_handler.retrieve_via_sql_query("item_id,item_name","inventory
")

    # finding current index

```

```

current_column_index = int(selected_item_id.get()[8:])

# in case at the last column

if current_column_index + 1 == len(table_information):
    current_column_index = 0
else:
    current_column_index = current_column_index + 1

selected_item_id.set("Item id: " +
str(table_information[current_column_index][0]-1))

set_selected_item_name()
set_basic_inventory_turnover()
set_reorder_warning()

def set_selected_order_name():
    # retriving the name list

    table_information =
database_handler.retrieve_via_sql_query("order_id,order_customer_name"
,"orders")

    # finding current index

    current_column_index = int(selected_order_id.get()[10:])

    # setting to name at the id

    set_orders_response_message("Order id set to: " +
str(table_information[current_column_index][0]-1))

    selected_order_name.set("Customer name: " +
table_information[current_column_index][1])

```



```

def select_previous_order():
    # retriving the name list
    table_information =
database_handler.retrieve_via_sql_query("order_id,order_customer_name"
,"orders")

    # finding current index
    current_column_index = int(selected_order_id.get()[10:])

    #in case at the first column
    if current_column_index == 0:
        current_column_index = len(table_information) - 1
    else:
        current_column_index = current_column_index - 1

    selected_order_id.set("Order id: " +
str(table_information[current_column_index][0]-1))

    set_selected_order_name()
    set_customer_spend_value()
    set_order_customer_number()

def select_next_order():
    # retriving the name list
    table_information =
database_handler.retrieve_via_sql_query("order_id","orders")

    # finding current index
    current_column_index = int(selected_order_id.get()[10:])

    # in case at the last column

```

```

if current_column_index + 1 == len(table_information):
    current_column_index = 0
else:
    current_column_index = current_column_index + 1

    selected_order_id.set("Order id: " +
str(table_information[current_column_index][0]-1))

    set_selected_order_name()
    set_customer_spend_value()
    set_order_customer_number()

# modelling view GUI
title_row = tk.Frame(modelling_view)
title_row.columnconfigure(0, weight = 1) # centering
title_row.columnconfigure(1, weight = 0)
title_row.columnconfigure(2, weight = 1)
title_row.grid(row = 0,
               colspan = 3,
               sticky = "nsew"
               )

inventory_view_button = Button(title_row,
                               text = "Inventory",
                               command =
switch_to_inventory_models_view,
                               bootstyle = "warning-outline"
                               )

```

```

inventory_view_button.grid(row = 0,
                           column = 0
                           )

order_view_button = Button(title_row,
                           text = "Orders",
                           command = switch_to_order_models_view,
                           bootstyle = "success-outline"
                           )
order_view_button.grid(row = 0,
                       column = 1
                       )

title_label = tk.Label(title_row, text = "
    


 Modelling
View 



", relief = "ridge", font = "TkFixedFont")
title_label.grid(row = 0,
                 column = 2
                 )

back_button = Button(title_row,
                     text = "Back",
                     command = switch_to_home_view,
                     bootstyle = "primary-outline"
                     )
back_button.grid(row = 0,
                 column = 3,
                 pady = 5
                 )

```

```

    )

inventory_models_view = tk.Frame(modelling_view)
inventory_models_view.grid(row = 1,
                           colspan = 2,
                           sticky = "nsew"
                           )

order_models_view = tk.Frame(modelling_view)
order_models_view.grid(row = 1,
                       colspan = 2,
                       sticky = "nsew"
                       )

# specific functions for inventory view
def get_selected_item_values():
    selected_item_values =
database_handler.retrieve_via_sql_query("item_cost,item_margin,item_stock,item_restock_value","inventory")
    return(selected_item_values)

def set_basic_inventory_turnover():
    selected_item_values = get_selected_item_values()
    # ^ in format [(item_name,item_cost,item_margin,item_stock), so on...]

    current_column_index = int(selected_item_id.get()[8:])

    cost = selected_item_values[current_column_index][0]

```

```

margin = selected_item_values[current_column_index][1]
stock = selected_item_values[current_column_index][2]

if stock == 0:
    basic_inventory_turnover_value.set("Basic inventory Turnover: 0.0")
else:
    basic_inventory_turnover_value.set("Basic inventory Turnover: "
+ str(round(((cost * (100 - margin)) / stock),6)))

def set_reorder_warning():
    selected_item_values = get_selected_item_values()
    # ^ in format [(item_name,item_cost,item_margin,item_stock), so
on...]

    current_column_index = int(selected_item_id.get()[8:])

    stock = selected_item_values[current_column_index][2]

    if stock < 100:
        reorder_warning_value.set("Reorder point reached: True")
    else:
        reorder_warning_value.set("Reorder point reached: False")

def populate_inventory_low_stocks_data():
    inventory_low_stocks_viewer_frame = tk.Frame(inventory_models_view)
    inventory_low_stocks_viewer_frame.grid(row = 1,
                                           column = 2,
                                           rowspan = 6,

```

```

        sticky = "e"
    )

    inventory_low_stocks_data =
database_handler.retrieve_via_sql_query("item_name,item_stock,item_res
tock_value","inventory")

    inventory_low_stocks_viewer =
Treeview(inventory_low_stocks_viewer_frame,

        columns =
("item_name","item_stock_restock_value_difference"),

        show = 'headings',
        height = 13,
        bootstyle = 'success'
    )

    inventory_low_stocks_viewer.grid(row = 1,

        column = 1
    )

    reformatted_data = []
    temp_inventory_low_stocks_data = inventory_low_stocks_data
    difference_list = []
    for i in inventory_low_stocks_data:
        # inventory_low_stocks_data in format
[(item_name,item_stock,item_restock_value)...so on]

        difference_list.append(i[1]-i[2]) # append the difference between
stock and restock value

    while len(difference_list) != 0:
        minimum_difference = min(difference_list)

```

```

        minimum_difference_index =
difference_list.index(minimum_difference)

        minimum_difference_data =
temp_inventory_low_stocks_data[minimum_difference_index]

reformatted_data.append((minimum_difference_data[0],minimum_difference
_data[1]-minimum_difference_data[2])) #appends a tuple in format
(name,difference)

        temp_inventory_low_stocks_data.pop(minimum_difference_index)
        difference_list.pop(minimum_difference_index)

# creating the scrollbar

        inventory_low_stocks_scrollbar =
ttk.Scrollbar(inventory_low_stocks_viewer_frame, orient = "vertical",
command = inventory_low_stocks_viewer.yview)

        inventory_low_stocks_scrollbar.grid(row = 1,
                                            column = 2,
                                            sticky = "nsew"
                                            )

        inventory_low_stocks_viewer.configure(yscrollcommand =
inventory_low_stocks_scrollbar.set)

# initialising columns

        inventory_low_stocks_viewer.column("item_name", anchor = "center",
width = 85)

        inventory_low_stocks_viewer.heading('item_name', text = 'Name')

inventory_low_stocks_viewer.column("item_stock_restock_value_differenc
e", anchor = "center", width = 55)

```



```

y_axis = tk.StringVar()
y_axis_column_name = tk.StringVar()
set_y_axis_button = tk.Button(inventory_models_view,
                               textvariable = y_axis,
                               command = set_y_axis
                               )

set_y_axis_button.grid(row = 2,
                       column = 0,
                       pady = 2,
                       sticky = "nsew"
                       )

y_axis_column_name.set("unspecified")
y_axis.set("Set Graph's Y-Axis:\n" + y_axis_column_name.get())


z_axis = tk.StringVar()
z_axis_column_name = tk.StringVar()
set_z_axis_button = tk.Button(inventory_models_view,
                               textvariable = z_axis,
                               command = set_z_axis
                               )

set_z_axis_button.grid(row = 3,
                       column = 0,
                       pady = 2,
                       sticky = "nsew"
                       )

z_axis_column_name.set("unspecified")
z_axis.set("Set Graph's Z-Axis:\n" + z_axis_column_name.get())

```

```

plot_button_frame = tk.Frame(inventory_models_view)
plot_button_frame.grid(row = 4,
                        column = 0,
                        sticky = "nsew"
                        )

scatter_plot_button = Button(plot_button_frame,
                             text = "
```

```

plot_type = tk.StringVar()
plot_type_button = tk.Button(inventory_models_view,
                             textvariable = plot_type,
                             command = toggle_plot_type
                             )

plot_type_button.grid(row = 5,
                      column = 0,
                      sticky = "nsew"
                      )

plot_type.set("📊 3D Graph type: Scatter")

date_frame = tk.Frame(inventory_models_view)
date_frame.grid(row = 6,
                column = 0,
                pady = 5,
                sticky = "nsew",
                )

day_month_year_label = tk.Label(date_frame, text = " Date | Month |
Year ", relief = "groove")
day_month_year_label.grid(row = 0,
                           columns = 3,
                           pady = 2,
                           sticky = "nsew"
                           )

current_date = datetime.date.today()

```

```

day_label = tk.Label(date_frame,
                      text = str(current_date)[8:],
                      relief = "raised",
                      width = 8
                      )

day_label.grid(row = 1,
               column = 0,
               padx = 1,
               sticky = "nsew"
               )


month_label = tk.Label(date_frame,
                       text = str(current_date)[5:7],
                       relief = "raised",
                       width = 8
                       )

month_label.grid(row = 1,
                 column = 1,
                 padx = 4,
                 sticky = "nsew"
                 )


year_label = tk.Label(date_frame,
                      text = str(current_date)[0:4],
                      relief = "raised",
                      width = 8
                      )

year_label.grid(row = 1,

```

```

        column = 2,
        padx = 1,
        sticky = "nsew"
    )

    # inventory column 2
    inventory_response_message = tk.StringVar()

    inventory_response_message_board = Label(inventory_models_view,
textvariable = inventory_response_message, bootstyle = "inverse-dark")

    inventory_response_message_board.grid(row = 0,
                                           column = 1,
                                           padx = 3,
                                           pady = 2,
                                           rowspan = 2,
                                           sticky = "nsew"
                                           )

    inventory_response_message.set("      Action status will be
displayed here:      \n\n")

    full_inventory_database_viewer_button = Button(inventory_models_view,
                                                    text = "View full
inventory database",
                                                    command =
tabularise_full_inventory_database,
                                                    bootstyle = "warning"
                                                    )

    full_inventory_database_viewer_button.grid(row = 2,
                                                column = 1,
                                                padx = 3,

```

```

        pady = 2,
        sticky = "nsew"
    )

    itemwise_statistics_frame = tk.Frame(inventory_models_view)
    itemwise_statistics_frame.grid(row = 3,
                                   column = 1
                                   )

    left_spacer = tk.Label(itemwise_statistics_frame, text = " ",
relief = "groove")
    left_spacer.grid(row = 0,
                     column = 0,
                     padx = 2
                     )

    itemwise_statistics_label = tk.Label(itemwise_statistics_frame,
                                         text = "Itemwise Statistics",
                                         relief = "groove"
                                         )

    itemwise_statistics_label.grid(row = 0,
                                   column = 1
                                   )

    right_spacer = tk.Label(itemwise_statistics_frame, text = " ",
relief = "groove")
    right_spacer.grid(row = 0,
                     column = 2,

```

```

        padx = 2
    )

previous_item_button = Button(itemwise_statistics_frame,
                              text = "<",
                              command = select_previous_item,
                              bootstyle = "success"
                              )

previous_item_button.grid(row = 1,
                          column = 0,
                          padx = 2
                          )

selected_item_id = tk.StringVar()
selected_item_label = tk.Label(itemwise_statistics_frame,
                               textvariable = selected_item_id,
                               relief = "groove"
                               )

selected_item_id.set("Item id: 0")
selected_item_label.grid(row = 1,
                        column = 1,
                        sticky = "nsew"
                        )

next_item_button = Button(itemwise_statistics_frame,
                           text = ">",
                           command = select_next_item,
                           bootstyle = "success"
                           )

```

```

        )

next_item_button.grid(row = 1,
                      column = 2,
                      padx = 2
                      )

selected_item_name = tk.StringVar()
selected_item_name_label = tk.Label(inventory_models_view,
                                   textvariable = selected_item_name,
                                   relief = "groove"
                                   )

set_selected_item_name()
selected_item_name_label.grid(row = 4,
                              column = 1,
                              pady = 5,
                              padx = 2,
                              sticky = "nsew"
                              )


basic_inventory_turnover_value = tk.StringVar()
basic_inventory_turnover_label = Label(inventory_models_view,
                                       textvariable =
basic_inventory_turnover_value,
                                       bootstyle = "inverse-dark")

set_basic_inventory_turnover()
basic_inventory_turnover_label.grid(row = 5,
                                    column = 1,
                                    pady = 1,

```



```

        padx = 3,
        sticky = "nsew"
    )

reorder_warning_value = tk.StringVar()
reorder_warning_label = Label(inventory_models_view,
                              textvariable =
reorder_warning_value,

                              bootstyle = "inverse-dark")

set_reorder_warning()
reorder_warning_label.grid(row = 6,
                           column = 1,
                           pady = 7,
                           padx = 3,
                           sticky = "nsew"
                           )

# inventory column 3
inventory_low_stocks_label = tk.Label(inventory_models_view, text =
"▣ Stocks Till Restock ▣", relief = "ridge")
inventory_low_stocks_label.grid(row = 0,
                                column = 2,
                                sticky = "nsew"
                                )

populate_inventory_low_stocks_data()

# wrapper functions for orders view
def orders_set_x_axis():
    set_x_axis(use_orders_table = True)

```

```

def orders_set_y_axis():
    set_y_axis(use_orders_table = True)

def orders_set_z_axis():
    set_z_axis(use_orders_table = True)

def orders_draw_plot():
    draw_plot(use_orders_table = True)

def orders_tabularise_full_database():
    tabularise_full_inventory_database(use_orders_table = True)

# specific functions for orders view
def set_customer_spend_value():

    current_order_id = int(selected_order_id.get()[9:]) + 1
    where_clause = "order_id =" + str(current_order_id)

    retrieved_data =
database_handler.retrieve_via_sql_query("order_id,order_final_cost,order_quantity", "orders", where_clause)

    #retrieved_data in format
    [(order_id,order_final_cost,order_quantity)]

    customer_spend_value.set("Total amt: " +
str(retrieved_data[0][1]*retrieved_data[0][2]))

def set_order_customer_number():

```

```

current_order_id = int(selected_order_id.get()[9:]) + 1
where_clause = "order_id =" + str(current_order_id)

retrieved_data =
database_handler.retrieve_via_sql_query("order_id,order_customer_contact_no","orders",where_clause)

#retrieved_data in format [(order_id,order_customer_contact_no)]
try:
    order_customer_number.set("Contact Number: " +
str(retrieved_data[0][1]))
except:
    # incase the customer didn't give their phone number
    order_customer_number.set("Contact Number: Not Available")

def populate_orders_highest_spends_data():
    orders_highest_spends_viewer_frame = tk.Frame(order_models_view)
    orders_highest_spends_viewer_frame.grid(row = 1,
                                             column = 2,
                                             rowspan = 6,
                                             sticky = "e"
                                             )

    orders_highest_spends_data =
database_handler.retrieve_via_sql_query("order_customer_name,sum(order
_final_cost*order_quantity) as total_spend","orders",sql_group_by =
"order_customer_name")

    orders_highest_spends_viewer =
Treeview(orders_highest_spends_viewer_frame,

```

```

        columns =
("order_customer_name", "total_spend"),

        show = 'headings',
        height = 13,
        bootstyle = 'success'
    )

orders_highest_spends_viewer.grid(row = 1,
                                  column = 1
                                  )

orders_reformatted_data = []
temp_orders_highest_spends_data = orders_highest_spends_data
total_spends_list = []
for i in orders_highest_spends_data:
    # orders_highest_spends_data in format
    [(order_customer_name, total_spend)...so on]

    total_spends_list.append(i[1]) # append the total spend

while len(total_spends_list) != 0:
    maximum_spend_index =
total_spends_list.index(max(total_spends_list))

    maximum_spend_data =
temp_orders_highest_spends_data[maximum_spend_index]

orders_reformatted_data.append((maximum_spend_data[0], maximum_spend_data[1])) #appends a tuple in format (name, total_spend)

temp_orders_highest_spends_data.pop(maximum_spend_index)

```

```

        total_spends_list.pop(maximum_spend_index)

        # creating the scrollbar

        orders_highest_spends_scrollbar =
ttk.Scrollbar(orders_highest_spends_viewer_frame, orient = "vertical",
command = orders_highest_spends_viewer.yview)

        orders_highest_spends_scrollbar.grid(row = 1,
                                             column = 2,
                                             sticky = "nsew"
                                             )

        orders_highest_spends_viewer.configure(yscrollcommand =
orders_highest_spends_scrollbar.set)

        # initialising columns

        orders_highest_spends_viewer.column("order_customer_name", anchor =
"center", width = 75)

        orders_highest_spends_viewer.heading('order_customer_name', text =
'Name')

        orders_highest_spends_viewer.column("total_spend", anchor =
"center", width = 65)

        orders_highest_spends_viewer.heading('total_spend', text = "Total")

        # insert values into orders_highest_spends_viewer

        for i in orders_reformatted_data:

            orders_highest_spends_viewer.insert(parent = '', index = tk.END,
values = i)

        # orders column 1

        orders_graph_plotter_label = tk.Label(order_models_view, text = "
Graph Plotter ", relief = "ridge")

        orders_graph_plotter_label.grid(row = 0,

```

```

        column = 0
    )

orders_x_axis = tk.StringVar()
orders_x_axis_column_name = tk.StringVar()
orders_set_x_axis_button = tk.Button(order_models_view,
                                     textvariable = orders_x_axis,
                                     command = orders_set_x_axis
                                     )
orders_set_x_axis_button.grid(row = 1,
                              column = 0,
                              pady = 2,
                              sticky = "nsew"
                              )

orders_x_axis_column_name.set("unspecified")
orders_x_axis.set("Set Graph's X-Axis:\n" +
orders_x_axis_column_name.get())


orders_y_axis = tk.StringVar()
orders_y_axis_column_name = tk.StringVar()
orders_set_y_axis_button = tk.Button(order_models_view,
                                     textvariable = orders_y_axis,
                                     command = orders_set_y_axis
                                     )
orders_set_y_axis_button.grid(row = 2,
                              column = 0,
                              pady = 2,
                              sticky = "nsew"
                              )

```

```

        )

orders_y_axis_column_name.set("unspecified")

orders_y_axis.set("Set Graph's Y-Axis:\n" +
orders_y_axis_column_name.get())

orders_z_axis = tk.StringVar()
orders_z_axis_column_name = tk.StringVar()
orders_set_z_axis_button = tk.Button(order_models_view,
                                     textvariable = orders_z_axis,
                                     command = orders_set_z_axis
                                     )

orders_set_z_axis_button.grid(row = 3,
                              column = 0,
                              pady = 2,
                              sticky = "nsew"
                              )

orders_z_axis_column_name.set("unspecified")

orders_z_axis.set("Set Graph's Z-Axis:\n" +
orders_z_axis_column_name.get())

orders_plot_button_frame = tk.Frame(order_models_view)
orders_plot_button_frame.grid(row = 4,
                              column = 0,
                              sticky = "nsew"
                              )

orders_scatter_plot_button = Button(orders_plot_button_frame,
                                    text = "
```

```

        bootstyle = "warning-outline"
    )

orders_scatter_plot_button.grid(row = 0,
                                column = 0,
                                padx = 2,
                                pady = 5
                                )

orders_graph_close_button = Button(orders_plot_button_frame,
                                    text = "Close Graph",
                                    command = close_plot,
                                    bootstyle = "success-outline"
                                    )

orders_graph_close_button.grid(row = 0,
                                column = 1,
                                padx = 2,
                                pady = 5
                                )

orders_plot_type = tk.StringVar()
orders_plot_type_button = tk.Button(order_models_view,
                                    textvariable = plot_type,
                                    command = toggle_plot_type
                                    )

orders_plot_type_button.grid(row = 5,
                              column = 0,
                              sticky = "nsew"
                              )

```



```

orders_plot_type.set("📈 3D Graph type: Scatter")

orders_date_frame = tk.Frame(order_models_view)
orders_date_frame.grid(row = 6,
                        column = 0,
                        pady = 5,
                        sticky = "nsew",
                        )

orders_day_month_year_label = tk.Label(orders_date_frame, text = "
Date | Month | Year ", relief = "groove")
orders_day_month_year_label.grid(row = 0,
                                columnspan = 3,
                                pady = 2,
                                sticky = "nsew"
                                )

orders_day_label = tk.Label(orders_date_frame,
                            text = str(current_date)[8:],
                            relief = "raised",
                            width = 8
                            )

orders_day_label.grid(row = 1,
                      column = 0,
                      padx = 1,
                      sticky = "nsew"
                      )

```

```

orders_month_label = tk.Label(orders_date_frame,
                               text = str(current_date)[5:7],
                               relief = "raised",
                               width = 8
                               )

orders_month_label.grid(row = 1,
                        column = 1,
                        padx = 4,
                        sticky = "nsew"
                        )

orders_year_label = tk.Label(orders_date_frame,
                              text = str(current_date)[0:4],
                              relief = "raised",
                              width = 8
                              )

orders_year_label.grid(row = 1,
                       column = 2,
                       padx = 1,
                       sticky = "nsew"
                       )

# orders column 2
orders_response_message = tk.StringVar()

orders_inventory_response_message_board = Label(order_models_view,
textvariable = orders_response_message, bootstyle = "inverse-dark")

orders_inventory_response_message_board.grid(row = 0,
                                              column = 1,

```

```

        padx = 3,
        pady = 2,
        rowspan = 2,
        sticky = "nsew"
    )

    orders_response_message.set("      Action status will be displayed
here:      \n\n")

    orders_full_inventory_database_viewer_button =
Button(order_models_view,

        text = "View full
orders database",

        command =

orders_tabularise_full_database,

        bootstyle = "success"
    )

    orders_full_inventory_database_viewer_button.grid(row = 2,

        column = 1,
        padx = 3,
        pady = 2,
        sticky = "nsew"
    )

    orderwise_statistics_frame = tk.Frame(order_models_view)
    orderwise_statistics_frame.grid(row = 3,

        column = 1
    )

```

```

orders_left_spacer = tk.Label(orderwise_statistics_frame, text = "
", relief = "groove")
orders_left_spacer.grid(row = 0,
                        column = 0,
                        padx = 2
                        )

orderwise_statistics_label = tk.Label(orderwise_statistics_frame,
                                     text = "■ ■ ■ Orderwise Statistics
■ ■ ■",
                                     relief = "groove"
                                     )
orderwise_statistics_label.grid(row = 0,
                                column = 1
                                )

orders_right_spacer = tk.Label(orderwise_statistics_frame, text = "
", relief = "groove")
orders_right_spacer.grid(row = 0,
                        column = 2,
                        padx = 2
                        )

orders_previous_item_button = Button(orderwise_statistics_frame,
                                     text = "<",
                                     command = select_previous_order,
                                     bootstyle = "success"
                                     )
orders_previous_item_button.grid(row = 1,

```

```

        column = 0,
        padx = 2
    )

selected_order_id = tk.StringVar()
selected_order_label = tk.Label(orderwise_statistics_frame,
                                textvariable = selected_order_id,
                                relief = "groove"
                                )

selected_order_id.set("Order id: 0")
selected_order_label.grid(row = 1,
                           column = 1,
                           sticky = "nsew"
                           )

orders_next_item_button = Button(orderwise_statistics_frame,
                                  text = ">",
                                  command = select_next_order,
                                  bootstyle = "success"
                                  )

orders_next_item_button.grid(row = 1,
                              column = 2,
                              padx = 2
                              )

selected_order_name = tk.StringVar()
selected_order_name_label = tk.Label(order_models_view,
                                      textvariable = selected_order_name,

```

```

        relief = "groove"
    )

    set_selected_order_name()
    selected_order_name_label.grid(row = 4,
                                   column = 1,
                                   pady = 5,
                                   padx = 2,
                                   sticky = "nsew"
    )

    customer_spend_value = tk.StringVar()
    customer_spend_label = Label(order_models_view,
                                textvariable =
customer_spend_value,
                                bootstyle = "inverse-dark")

    set_customer_spend_value()
    customer_spend_label.grid(row = 5,
                              column = 1,
                              pady = 1,
                              padx = 3,
                              sticky = "nsew"
    )

    order_customer_number = tk.StringVar()
    order_customer_number_label = Label(order_models_view,
                                       textvariable =
order_customer_number,
                                       bootstyle = "inverse-dark")

    set_order_customer_number()

```

```

order_customer_number_label.grid(row = 6,
                                column = 1,
                                pady = 7,
                                padx = 3,
                                sticky = "nsew"
                                )

# orders column 3
inventory_low_stocks_label = tk.Label(order_models_view, text = " ▣
Highest Spenders ▣ ", relief = "ridge")
inventory_low_stocks_label.grid(row = 0,
                                column = 2,
                                sticky = "nsew"
                                )

populate_orders_highest_spends_data()

inventory_models_view.tkraise()

# Sets initial frame to be home_view
home_view.tkraise()

root.mainloop()

```

[Module Overview]

The `visual_handler.py` module provides a Graphical User Interface (GUI) for an Inventory and Order Analytics System.

It integrates database operations, file import, and visualization to help users manage and analyze inventory/order datasets.

[Main Function]

run_GUI()

Description: Entry point of the GUI.

Responsibilities:

- Initialize Tkinter root window (Tk() object).
- Configure Home and Modelling views.
- Create styled UI components using TTK Bootstrap.
- Define all nested functions for database operations, plotting, and navigation.
- Run the Tkinter event loop (mainloop).

[Home View Functions]

Function	Purpose	Key Details
<i>populate_data()</i>	<i>Displays inventory and orders tables in home view.</i>	<i>Uses Tkinter Treeview with vertical scrollbars; fetches data via <code>database_handler.retrieve_via_sql_query()</code>.</i>
<i>import_database()</i>	<i>Imports CSV data for inventory and orders.</i>	<i>Uses NumPy's <code>genfromtxt</code>. Calls <code>database_handler.import_items()</code> and <code>database_handler.import_orders()</code>. Handles errors with messages.</i>
<i>set_inventory_path() / set_orders_path()</i>	<i>Opens file dialogs to select CSV files.</i>	<i>Updates StringVar labels with truncated file paths for clarity.</i>
<i>refresh()</i>	<i>Refreshes database displays and model statistics.</i>	<i>Calls <code>populate_data()</code>, <code>populate_inventory_low_stocks_data()</code>, and <code>populate_orders_highest_spends_data()</code>.</i>

<code>switch_to_modelling_view()</code>	Switches GUI to Modelling View.	Uses <code>tkraise()</code> to bring the modelling frame to front.
---	---------------------------------	--

[Modelling View Functions]

Function	Purpose	Key Details
<code>switch_to_home_view()</code>	Returns to Home View.	Uses <code>tkraise()</code> .
<code>switch_to_inventory_mode ls_view() / switch_to_order_models_v iew()</code>	Switch between inventory/order analytics sections.	
<code>set_inventory_response_m essage() / set_orders_response_mess age()</code>	Updates status boards.	Maintains last 2 messages in a rotating buffer.
<code>close_plot()</code>	Closes all open Matplotlib plots.	Calls <code>plt.close()</code> .
<code>draw_plot(use_orders_tab le=False)</code>	Creates 2D/3D plots.	- Retrieves selected x, y, z columns. - Supports both string and numeric axes. - For 3D: supports Scatter and Line plots.
<code>toggle_plot_type()</code>	Switches between Scatter and Line plot modes.	Updates message boards and button label.
<code>set_x_axis() / set_y_axis() / set_z_axis()</code>	Cycles through available database columns for axes.	Maintains current column index and loops back if at end.
<code>tabularise_full_inventor y_database(use_orders_ta ble=False)</code>	Opens new window to display full database table.	- Uses Treeview with scrollbars. - Supports both inventory and orders.

[Inventory-Specific Functions]

Function	Purpose	Key Details
<code>get_selected_item_values()</code>	Retrieves selected item details.	Queries <code>item_cost</code> , <code>item_margin</code> , <code>item_stock</code> , <code>item_restock_value</code> .
<code>set_basic_inventory_turnover()</code>	Calculates turnover ratio.	Formula: $(\text{cost} \times (100 - \text{margin})) / \text{stock}$.
<code>set_reorder_warning()</code>	Displays reorder warning.	Alerts when <code>stock < 100</code> .
<code>populate_inventory_low_stocks_data()</code>	Lists items closest to restocking point.	Sorts by $(\text{stock} - \text{restock_value})$ difference.
<code>set_selected_item_name()</code>	Displays current item name.	Syncs with <code>selected_item_id</code> .
<code>select_previous_item()</code> / <code>select_next_item()</code>	Navigate inventory items sequentially.	Wraps around at list edges.

[Order-Specific Functions]

Function	Purpose	Key Details
<code>set_selected_order_name()</code>	Displays customer name for selected order.	Syncs with <code>selected_order_id</code> .
<code>select_previous_order()</code> / <code>select_next_order()</code>	Navigate through orders sequentially.	Wraps around at start/end.

<code>set_customer_spend_value()</code>	<i>Calculates total spend for an order.</i>	<i>Formula: order_final_cost × order_quantity.</i>
<code>set_order_customer_number()</code>	<i>Displays customer contact number.</i>	<i>Handles missing phone numbers gracefully.</i>
<code>populate_orders_highest_spends_data()</code>	<i>Lists customers sorted by spend.</i>	<i>Aggregates data with SUM(order_final_cost × quantity) and sorts descending.</i>
<code>orders_set_x_axis()</code> / <code>orders_set_y_axis()</code> / <code>orders_set_z_axis()</code>	<i>Axis selection wrappers for order tables.</i>	<i>Calls generic axis functions with use_orders_table=True.</i>
<code>orders_draw_plot()</code>	<i>Plots order data.</i>	<i>Wrapper for draw_plot() with use_orders_table=True.</i>
<code>orders_tabularise_full_database()</code>	<i>Opens full orders table in new window.</i>	<i>Similar to inventory tabularization.</i>

Database Overview



main_database.orders

Table Details

Engine:	InnoDB
Row format:	Dynamic
Column count:	10
Table rows:	5
AVG row length:	3276
Data length:	16.0 KiB
Index length:	0.0 bytes
Max data length:	0.0 bytes
Data free:	0.0 bytes
Table size (estimate):	16.0 KiB
File format:	
Data path:	/usr/local/mysql/data/main_database/orders.ibd
Update time:	2025-08-25 22:28:49

Information on this page may be outdated. Click [Analyze](#) to update it.

Column	Type	Default Value	Nullable	Character Set	Collation
♦ order_id	int		NO		
♦ order_item_name	varchar(20)		NO	utf8mb4	utf8mb4_090...
♦ order_date	date		NO		
♦ order_initial_cost	float(8,2)		NO		
♦ order_gst	float(8,2)		NO		
♦ order_discount	float(5,2)		NO		
♦ order_final_cost	float(8,2)		NO		
♦ order_quantity	int		NO		
♦ order_customer_n...	varchar(20)		NO	utf8mb4	utf8mb4_090...
♦ order_customer_c...	varchar(10)		YES	utf8mb4	utf8mb4_090...

Indexes in Table

Visible	Key	Type	Uniq...	Columns
☒	 PRIMARY	BTREE	YES	order_id

Index Details

Key Name:
Index Type:
Allows NULL:
Cardinality:
Comment:
User Comment:

Columns in table

Column	Type	Nullable	Indexes
♦ order_id	int	NO	PRIMARY
♦ order_item_name	varchar(20)	NO	
♦ order_date	date	NO	
♦ order_initial_cost	float(8,2)	NO	
♦ order_gst	float(8,2)	NO	
♦ order_discount	float(5,2)	NO	
♦ order_final_cost	float(8,2)	NO	
♦ order_quantity	int	NO	



main_database.inventory

Table Details

Engine:	InnoDB
Row format:	Dynamic
Column count:	12
Table rows:	5
AVG row length:	3276
Data length:	16.0 KiB
Index length:	0.0 bytes
Max data length:	0.0 bytes
Data free:	0.0 bytes
Table size (estimate):	16.0 KiB
File format:	
Data path:	/usr/local/mysql/data/main_database/inventory.ibd
Update time:	2025-08-25 22:28:49

Information on this page may be outdated. Click [Analyze](#) to update it.

Column	Type	Default Value	Nullable	Character Set	Collation
◆ item_id	int		NO		
◆ item_name	varchar(20)		NO	utf8mb4	utf8mb4_090...
◆ item_cost	float(8,2)		NO		
◆ item_gst	float(8,2)		NO		
◆ item_discount	float(5,2)		NO		
◆ item_final_cost	float(8,2)		NO		
◆ item_margin	float(5,2)		NO		
◆ item_stock	int		NO		
◆ item_restock_value	int		NO		
◆ item_manufacture...	varchar(40)		NO	utf8mb4	utf8mb4_090...
◆ item_manufacture...	varchar(20)		NO	utf8mb4	utf8mb4_090...
◆ item_manufacture...	varchar(10)		NO	utf8mb4	utf8mb4_090...

Indexes in Table

Visible	Key	Type	Uniq...	Columns
☒	 PRIMARY	BTREE	YES	item_id

Index Details

Key Name:

Index Type:

Allows NULL:

Cardinality:

Comment:

User Comment:

Columns in table

Column	Type	Nullable	Indexes
 item_id	int	NO	PRIMARY
 item_name	varchar(20)	NO	
 item_cost	float(8,2)	NO	
 item_gst	float(8,2)	NO	
 item_discount	float(5,2)	NO	
 item_final_cost	float(8,2)	NO	
 item_margin	float(5,2)	NO	
 item_stock	int	NO	

GUI Overview

```
Please enter your MySQL username:
Please enter your MySQL password:
Please enter MySQL host: ('localhost' if using localhost):
Please enter MySQL port: ('n' if using localhost):
```

Python Inventory Analytics

Database Viewer

Inventory Orders

S.No	Name	Cost	Total	Stock
1	Apples	139.00	122.55	500
2	Raspberry	139.00	122.55	500
3	Orange	139.00	122.55	500
4	Bannana	139.00	122.55	500
5	Potato	139.00	122.55	500

Other Functions

Modelling Viewport

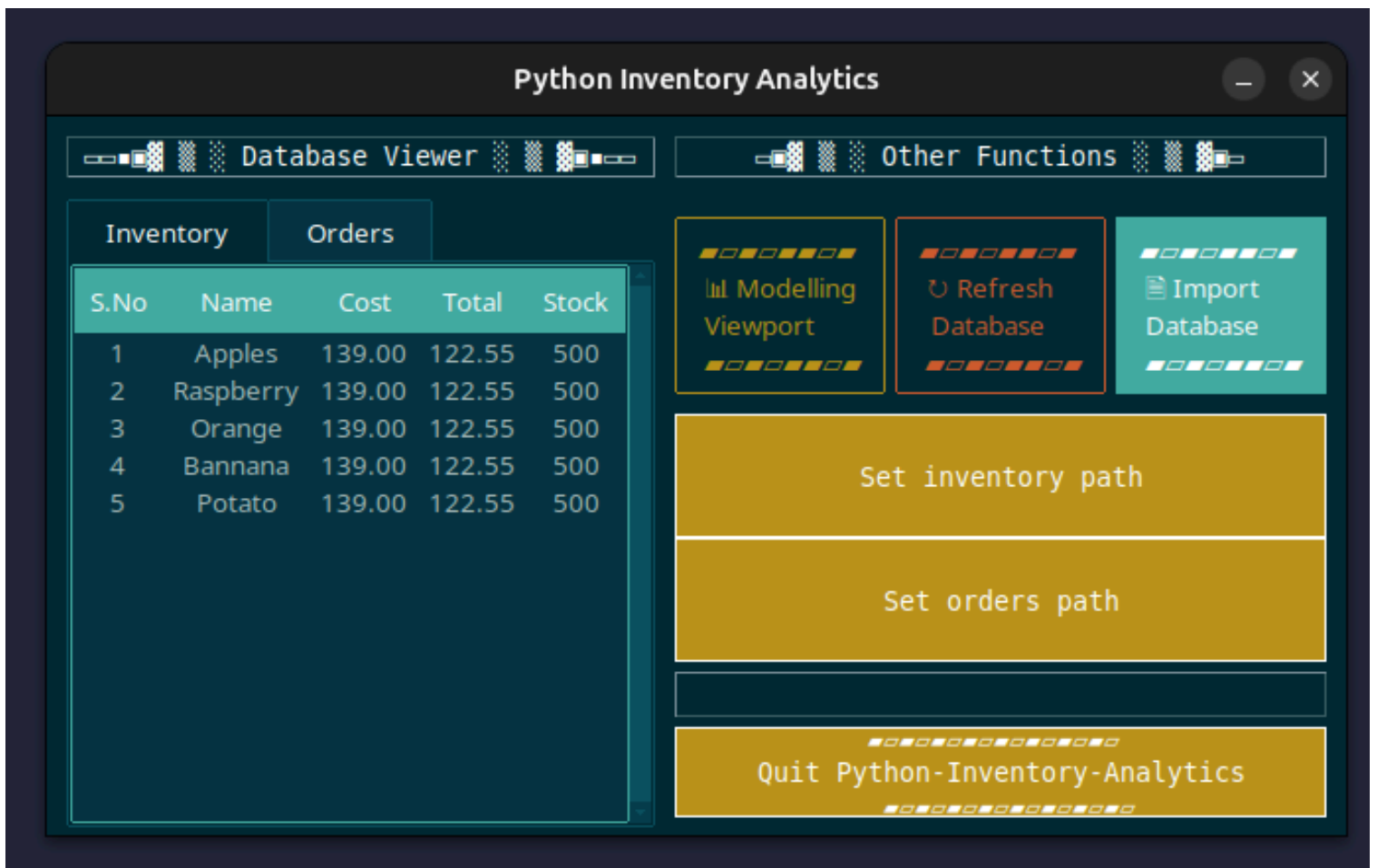
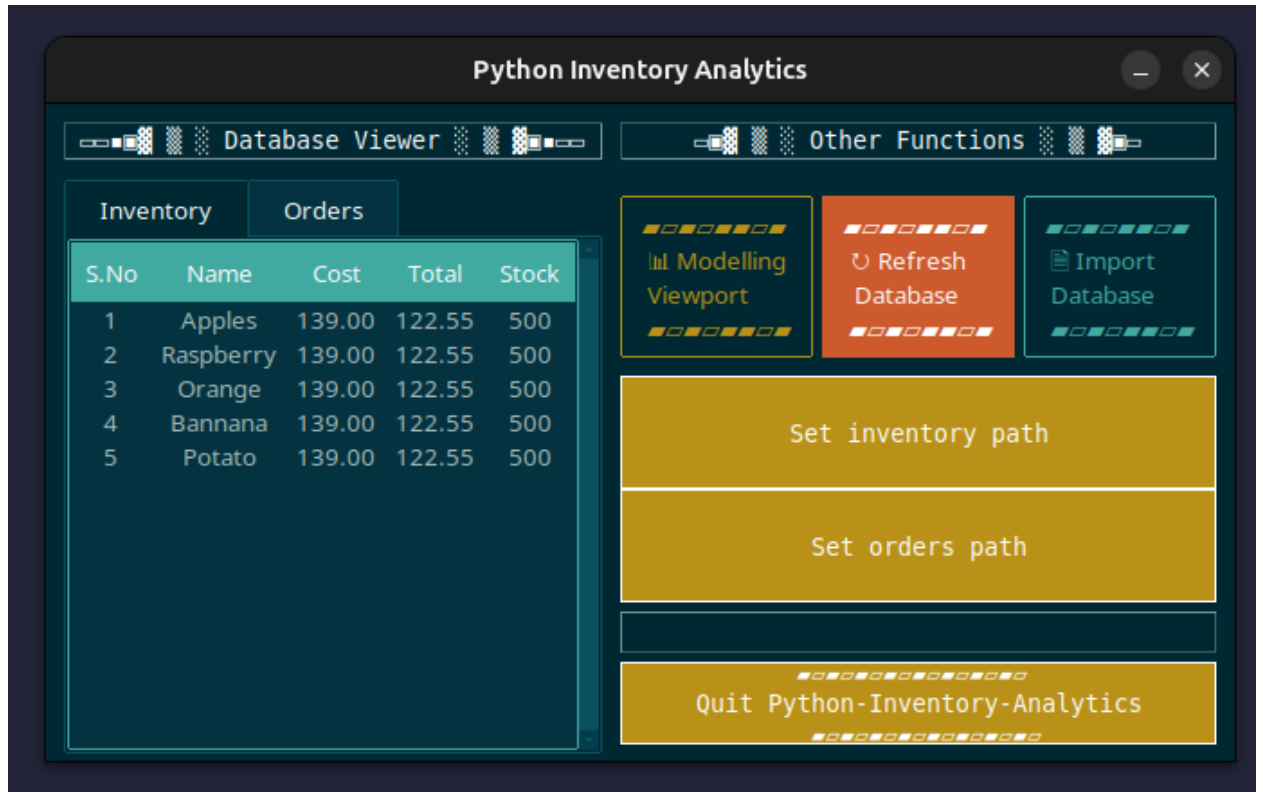
Refresh Database

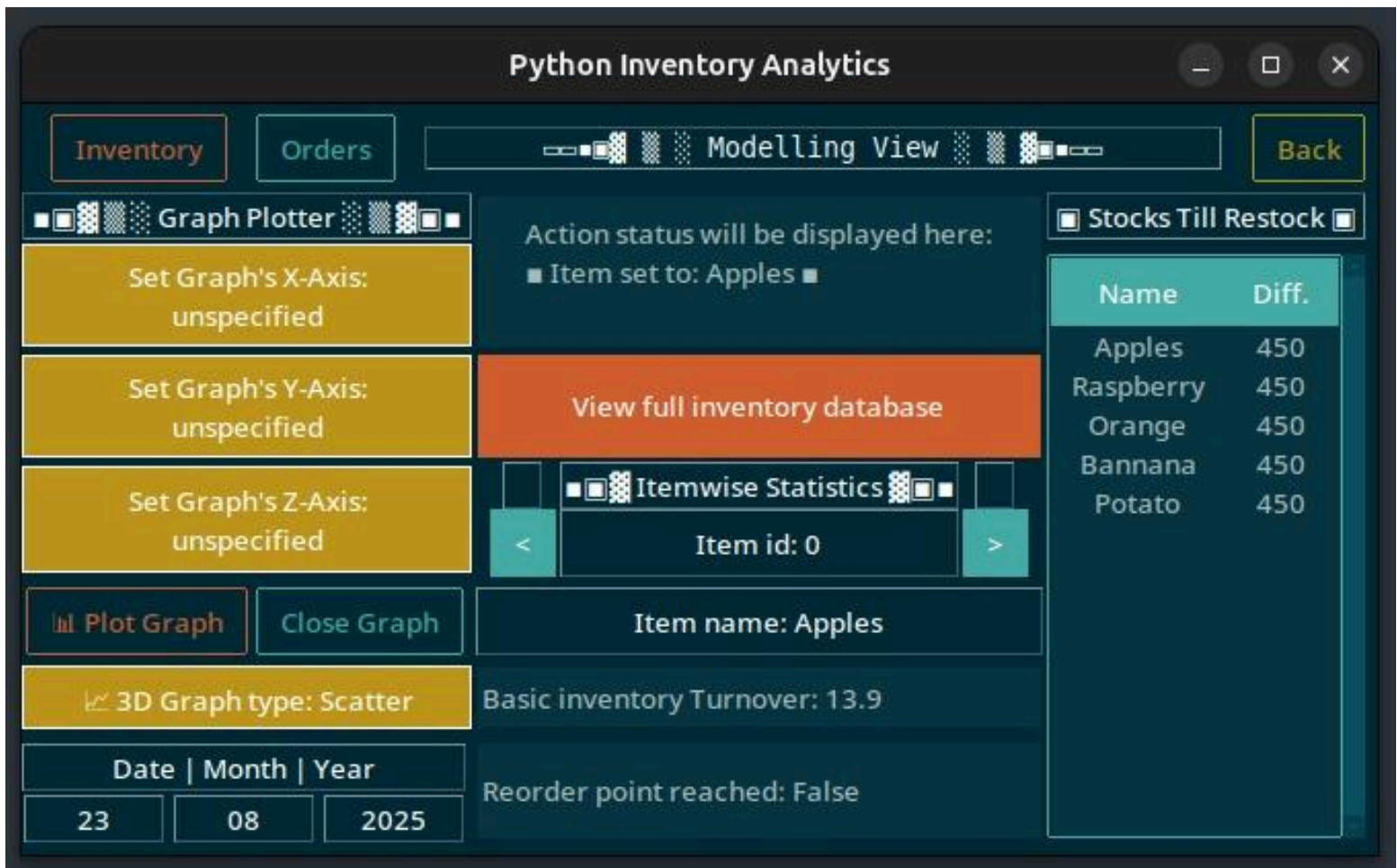
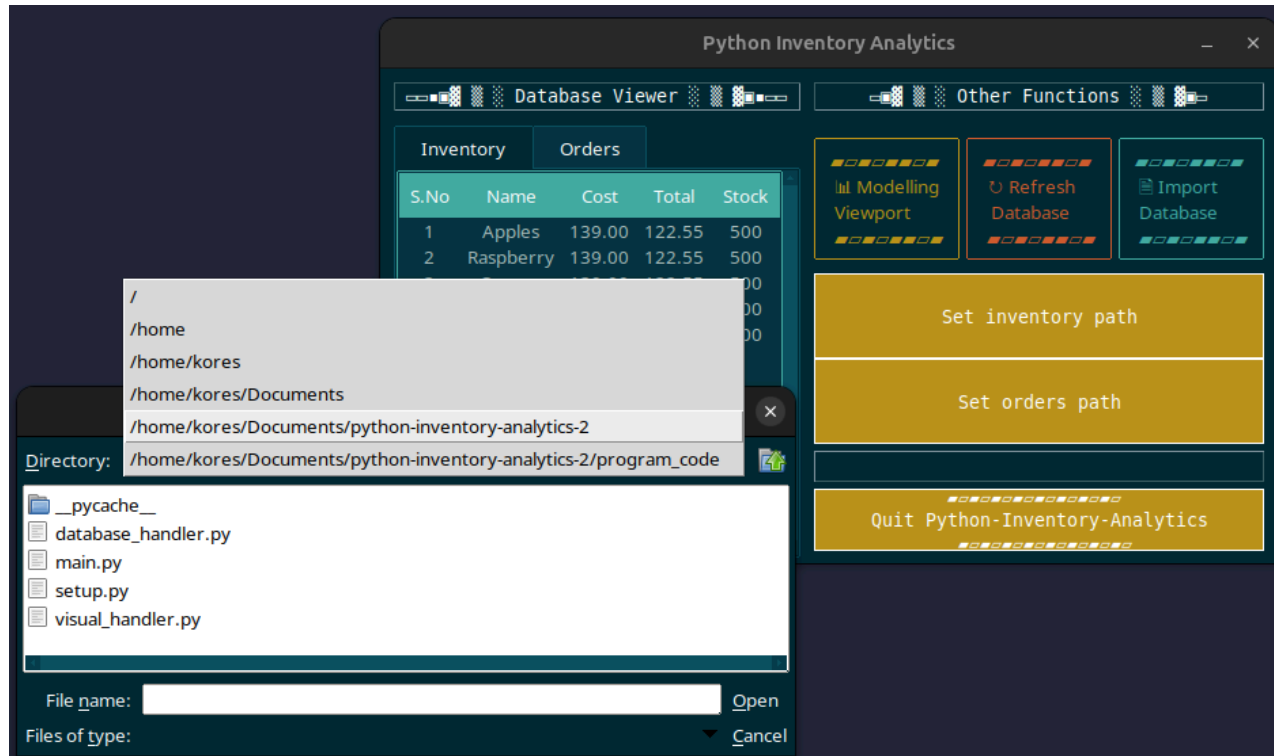
Import Database

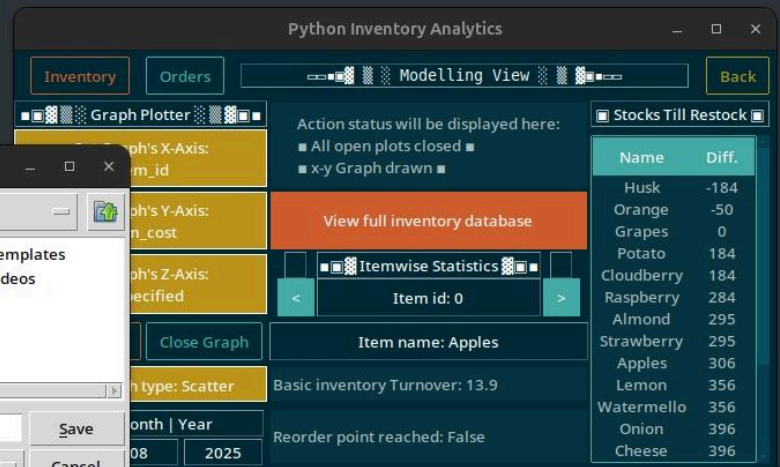
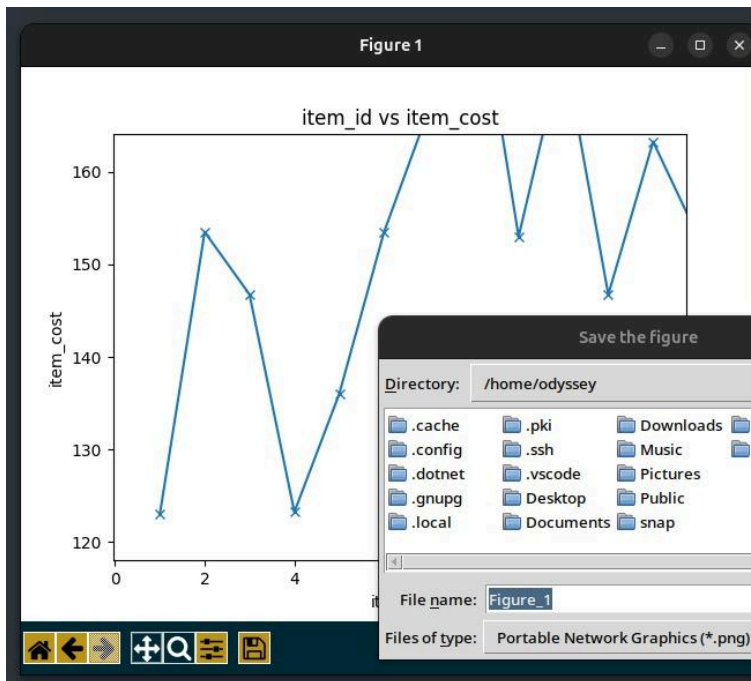
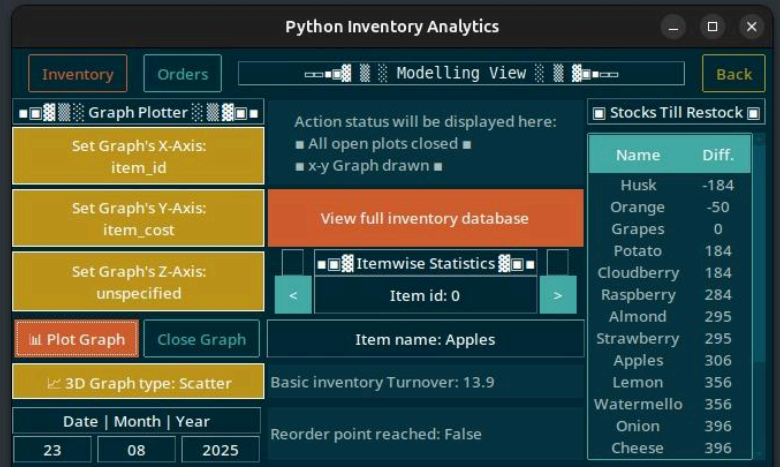
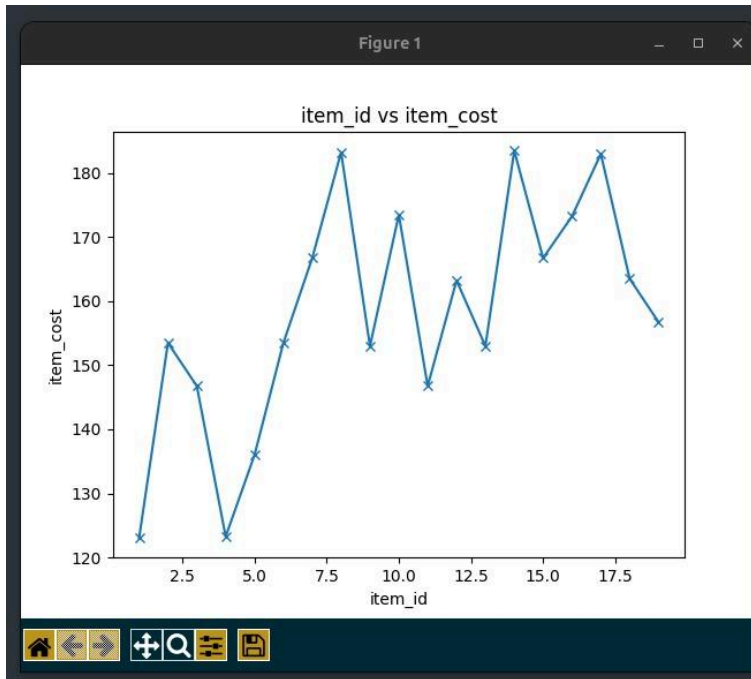
Set inventory path

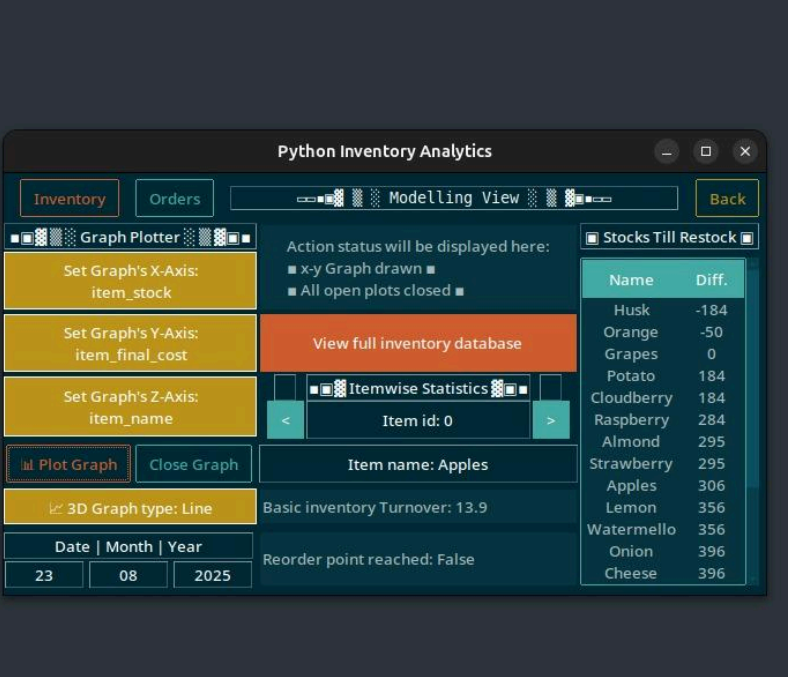
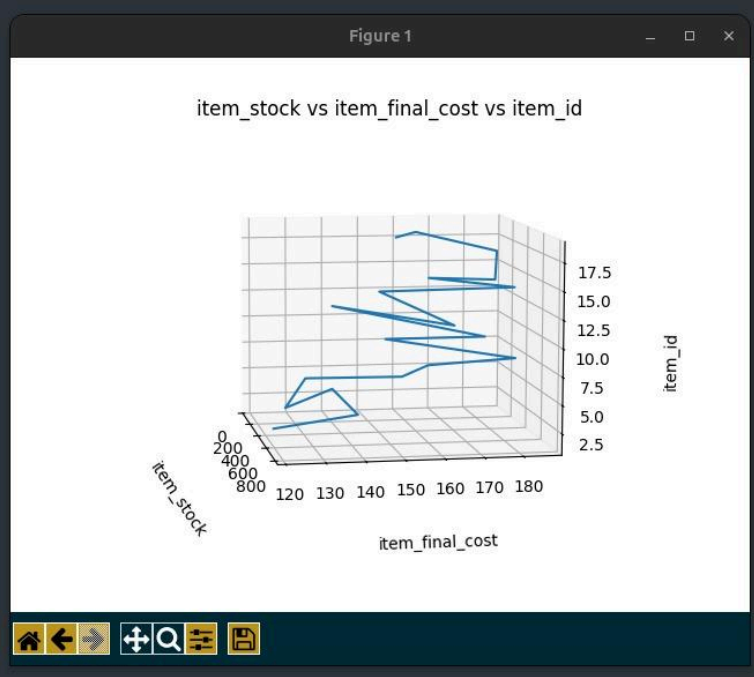
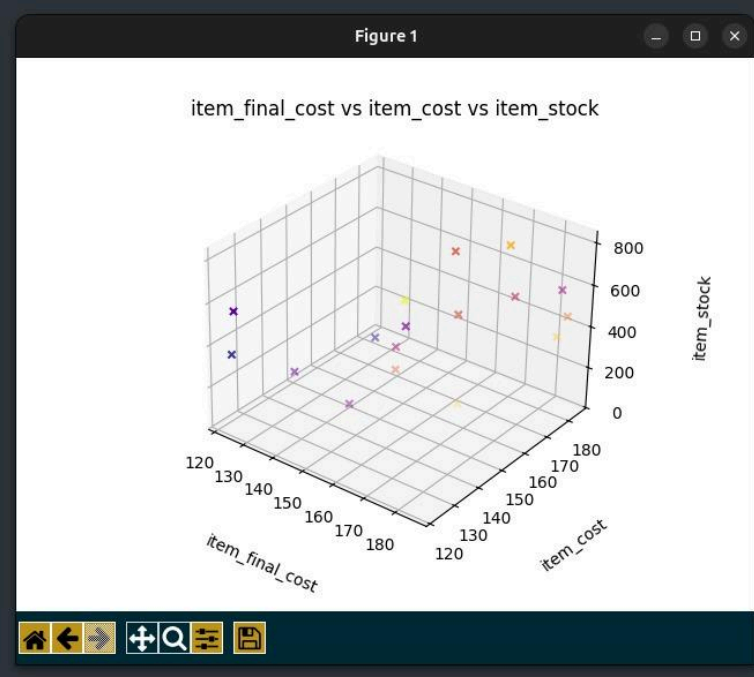
Set orders path

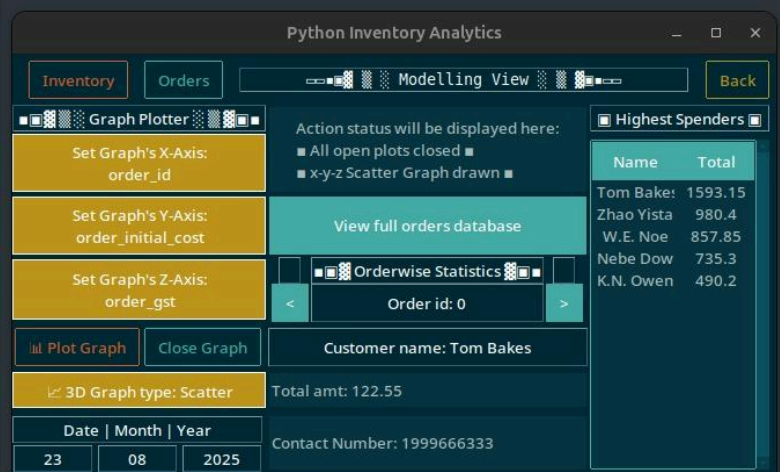
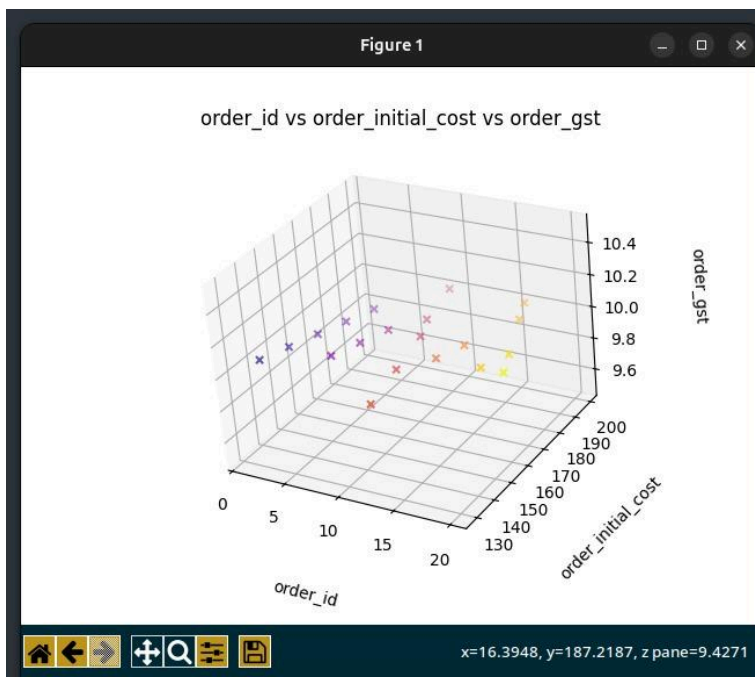
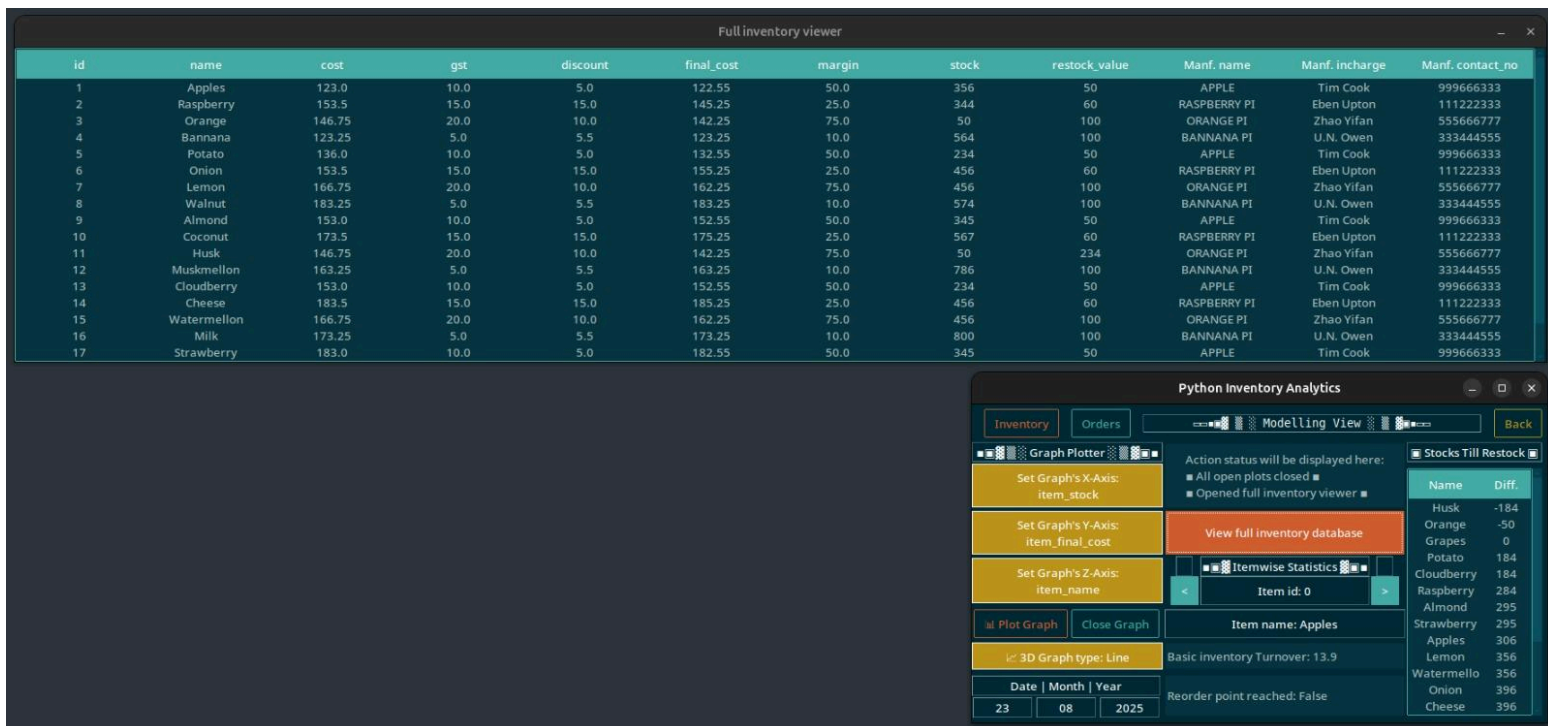
Quit Python-Inventory-Analytics

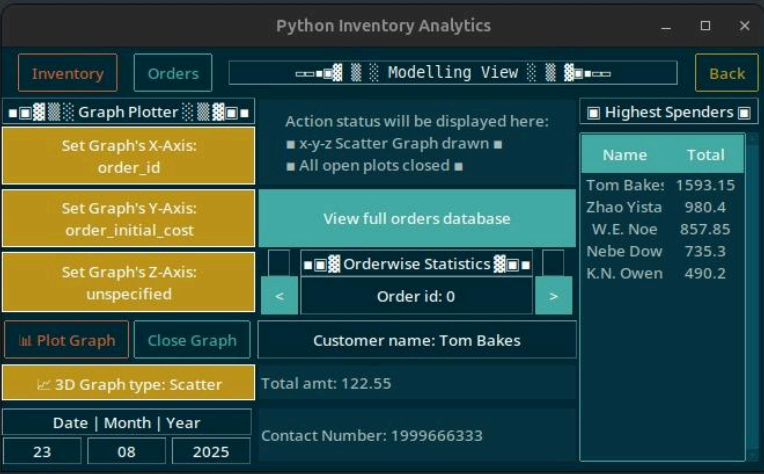
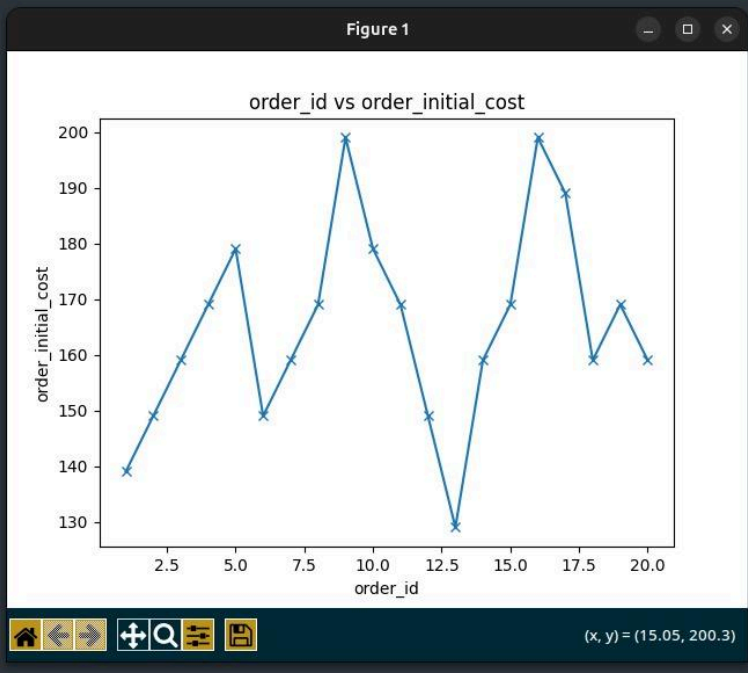












Full Orders viewer

id	item_name	date	initial_cost	gst	discount	final_cost	quantity	customer_name	customer_contact_no
1	Apples	2025-04-08	139.0	10.0	5.0	122.55	2	Tom Bakes	1999666333
2	Raspberry	2025-05-08	149.0	10.0	5.0	122.55	1	Nebe Downton	1111222333
3	Orange	2025-04-07	159.0	10.0	5.0	122.55	5	Zhao Yistan	1555666777
4	Bannana	2025-05-09	169.0	10.0	5.0	122.55	1	K.N. Owen	1333444555
5	Potato	2025-03-08	179.0	10.0	5.0	122.55	1	W.E. Noe	1777888999
6	Apples	2025-04-10	149.0	10.0	5.0	122.55	1	Tom Bakes	1999666333
7	Raspberry	2025-01-08	159.0	10.0	5.0	122.55	3	Nebe Downton	1111222333
8	Orange	2025-02-05	169.0	10.0	5.0	122.55	1	Zhao Yistan	1555666777
9	Bannana	2025-07-08	199.0	10.0	5.0	122.55	1	K.N. Owen	1333444555
10	Potato	2025-03-09	179.0	10.0	5.0	122.55	4	W.E. Noe	1777888999
11	Apples	2025-06-04	169.0	10.0	5.0	122.55	1	Tom Bakes	1999666333
12	Raspberry	2025-06-04	149.0	10.0	5.0	122.55	1	Nebe Downton	1111222333
13	Orange	2025-09-05	129.0	10.0	5.0	122.55	1	Zhao Yistan	1555666777
14	Bannana	2025-08-08	159.0	10.0	5.0	122.55	1	K.N. Owen	1333444555
15	Potato	2025-03-06	169.0	10.0	5.0	122.55	1	W.E. Noe	1777888999
16	Apples	2025-04-03	199.0	10.0	5.0	122.55	9	Tom Bakes	1999666333
17	Raspberry	2025-06-09	189.0	10.0	5.0	122.55	1	Nebe Downton	1111222333

